



Beyond Binary: Four-Class Risk Stratification from Gastrointestinal Endoscopy Using Asymmetric-Cost Lightweight CNN-Transformer Learning

DenseNet-121 · EfficientNet-B0 · DeiT-Tiny | Asymmetric Endoscopy Loss | GradCAM | MC Dropout

Author: Azizur Rahman

Affiliation: Indiana Wesleyan University · RadTH Technologies

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Notebook Contents

#	Section
1	Environment Setup
2	Dataset Loading & 4-Class Label Mapping
3	Data Augmentation & DataLoader
4	Model Architecture — DenseNet-121 · EfficientNet-B0 · DeiT-Tiny
5	Asymmetric Endoscopy Loss (AEL)
6	Training (one cell per model, checkpoint-safe)
7	AEL Ablation Study (single-seed reference run)
8	Cross-Dataset Evaluation — Kvasir-v2 zero-shot
9	ROC-AUC Curves
10	GradCAM Explainability
11	Monte Carlo Dropout Uncertainty
12	Endoscopist Workload Simulation
13	Multi-Seed Ablation — AEL vs CE vs Focal (Table 12)
14	Multi-Seed AEL Sensitivity — HR Weight Sweep (Table 11)
15	Model Calibration — ECE & Reliability Diagrams
16	Results Summary & Paper Tables

To reproduce paper results end-to-end:

1. Download HyperKvasir labeled-images → `data/HyperKvasir/labeled-images/`
2. Download Kvasir-v2 → `data/Kvasir-v2/`
3. Run cells top-to-bottom (each section is independent; checkpoints are saved after training)
4. Sections 13–14 take ~2–4 h each on MPS/GPU — results print LaTeX rows for Tables 11 & 12

1. Environment Setup

```
In [1]: !pip install torch torchvision timm
!pip install scikit-learn pandas numpy matplotlib seaborn tqdm
!pip install grad-cam opencv-python pillow
!pip install nbformat
```

Requirement already satisfied: torch in C:\Program Files\Python312\Lib\site-packages (2.7.1+cu118)

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 Requirement already satisfied: typing-extensions>=4.4.0 in C:\Program Files\Python312\Lib\site-packages (from referencing>=0.28.4->jsonschema>=2.6->nbformat) (4.15.0)

```

In [2]: import os as _os
PROJECT_DIR = _os.path.dirname(_os.path.abspath('GastroEndoscopy_Risk_Stratification'))
CKPT_DIR = _os.path.join(PROJECT_DIR, 'checkpoints')
print(f'Project dir: {PROJECT_DIR}')

import os, random, warnings, copy
import numpy as np
import pandas as pd
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader, WeightedRandomSampler
  
```

```

from torch.optim import AdamW
from torch.optim.lr_scheduler import CosineAnnealingLR
from torchvision import transforms, models
import timm
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    classification_report, f1_score, roc_auc_score,
    confusion_matrix, recall_score, roc_curve, auc
)
from sklearn.preprocessing import label_binarize
from PIL import Image
import matplotlib.pyplot as plt
import seaborn as sns
import cv2
from tqdm import tqdm
warnings.filterwarnings('ignore')

# — Reproducibility —
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed_all(SEED)
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False

# — Device —
if torch.cuda.is_available():
    DEVICE = torch.device('cuda')
elif torch.backends.mps.is_available():
    DEVICE = torch.device('mps')
else:
    DEVICE = torch.device('cpu')

# — Global hyperparameters —
NUM_CLASSES = 4
IMG_SIZE = 224
BATCH_SIZE = 32

# — CLAHE config —
CLAHE_CLIP_LIMIT = 2.0
CLAHE_TILE_SIZE = (8, 8)

# — AEL class weights (ACG/ESGE clinical cost asymmetry) —
# Inflammatory raised 2.0 → 3.5 to address small class size (645 images)
AEL_WEIGHTS = [1.0, 3.5, 3.0, 5.0] # Normal, Inflammatory, Pre-malignant, Hi

# — Per-model dropout rates —
DROPOUT_RATES = {
    'densenet121': 0.5,
    'efficientnet_b0': 0.3,
    'deit_tiny': 0.1,
}

```

```

LABEL_NAMES = ['Normal', 'Inflammatory', 'Pre_malignant', 'High_Risk']
RISK_LABELS = ['Routine Surveillance', 'Medical Management',
               'Biopsy + Surveillance', 'Immediate Intervention']
RISK_ACTIONS = [
    'Continue standard screening interval',
    'Medical treatment + annual endoscopy',
    'Biopsy required + 3-6 month follow-up',
    'Resection or oncology referral',
]

print(f'Device : {DEVICE}')
print(f'PyTorch: {torch.__version__}')
print(f'Seeds : SEED={SEED} cudnn.deterministic=True')
print(f'AEL weights: {AEL_WEIGHTS} (Inflammatory raised to 3.5)')
print(f'\n4-Class Clinical Risk Schema (ACG/ESGE Aligned):')
for i in range(NUM_CLASSES):
    print(f' Class {i} | {LABEL_NAMES[i]:<18} → {RISK_LABELS[i]}')

```

```

Project dir: D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification
Device : cuda
PyTorch: 2.7.1+cu118
Seeds : SEED=42 cudnn.deterministic=True
AEL weights: [1.0, 3.5, 3.0, 5.0] (Inflammatory raised to 3.5)

```

```

4-Class Clinical Risk Schema (ACG/ESGE Aligned):
Class 0 | Normal          → Routine Surveillance
Class 1 | Inflammatory    → Medical Management
Class 2 | Pre_malignant   → Biopsy + Surveillance
Class 3 | High_Risk       → Immediate Intervention

```

2. Dataset Loading & 4-Class Label Mapping

HyperKvasir contains 110,079 images across 23 GI finding classes. We map these to 4 clinical risk tiers following ACG/ESGE guidelines.

Risk Class	HyperKvasir Folders
0 Normal	normal-cecum, normal-pylorus, normal-z-line, retroflex-stomach, retroflex-rectum, ileum, bbps-2-3
1 Inflammatory	esophagitis-a, ulcerative-colitis-grade-0-1, ulcerative-colitis-grade-1, hemorrhoids
2 Pre-malignant	barretts, barretts-short-segment, esophagitis-b-d, polyp, ulcerative-colitis-grade-1-2, ulcerative-colitis-grade-2
3 High-Risk	ulcerative-colitis-grade-2-3, ulcerative-colitis-grade-3, dyed-lifted-polyps, dyed-resection-margins

```

In [3]: # Actual HyperKvasir folder names (verified from extracted archive)
        HYPERKVASIR_MAP = {
            # Class 0 – Normal / Routine Surveillance

```

```

'cecum':          0, # lower-gi-tract/anatomical-landmarks
'pylorus':        0, # upper-gi-tract/anatomical-landmarks
'z-line':         0, # upper-gi-tract/anatomical-landmarks
'retroflex-stomach': 0,
'retroflex-rectum': 0,
'ileum':          0,
'bbps-2-3':       0,
# Class 1 – Inflammatory / Low-Risk
'esophagitis-a':  1,
'ulcerative-colitis-grade-0-1': 1,
'ulcerative-colitis-grade-1':  1,
'hemorrhoids':    1,
# Class 2 – Pre-malignant / Moderate-Risk
'barretts':       2,
'barretts-short-segment': 2,
'esophagitis-b-d': 2,
'polyps':         2, # actual folder name is 'polyps' not
'ulcerative-colitis-grade-1-2': 2,
'ulcerative-colitis-grade-2':  2,
# Class 3 – High-Risk / Immediate Intervention
'ulcerative-colitis-grade-2-3': 3,
'ulcerative-colitis-grade-3':  3,
'dyed-lifted-polyps':          3,
'dyed-resection-margins':      3,
}

EXCLUDED_CLASSES = {'bbps-0-1', 'impacted-stool', 'out-of-patient', 'short-seg

def load_hyperkvasir(data_dir: str, max_normal: int = 3000) -> pd.DataFrame:
    """
    Walk the nested HyperKvasir structure:
    labeled-images/<tract>/<category>/<class-folder>/<images>
    and map every class folder to a 4-class clinical risk tier.
    """
    rng = random.Random(SEED)
    base_dir = os.path.join(data_dir, 'labeled-images')

    if not os.path.exists(base_dir):
        print(f'[!] HyperKvasir not found at {base_dir}')
        return pd.DataFrame(columns=['image_path', 'label', 'label_name', 'risk_tier'])

    normal_paths, other_rows, skipped = [], [], []

    # Walk all subdirectories – class folders sit at depth 3 (tract/category/class)
    for root, dirs, files in os.walk(base_dir):
        depth = root.replace(base_dir, '').count(os.sep)
        if depth != 3:
            continue
        class_folder = os.path.basename(root)
        if class_folder in EXCLUDED_CLASSES:
            continue
        label = HYPERKVASIR_MAP.get(class_folder)

```

```

        if label is None:
            skipped.append(class_folder)
            continue
        imgs = [os.path.join(root, f) for f in files
                 if f.lower().endswith(('.jpg', '.jpeg', '.png', '.bmp'))]
        for path in imgs:
            row = {'image_path': path, 'label': label,
                  'label_name': LABEL_NAMES[label], 'risk': RISK_LABELS[label],
                  'source_class': class_folder}
            (normal_paths if label == 0 else other_rows).append(row)

    if skipped:
        print(f'[INFO] Skipped {len(skipped)} unmapped folders: {"", ".join(sorted(skipped))}')

    if len(normal_paths) > max_normal:
        print(f'[INFO] Normal class capped: {len(normal_paths)} → {max_normal}')
        rng.shuffle(normal_paths)
        normal_paths = normal_paths[:max_normal]

    df = pd.DataFrame(normal_paths + other_rows)
    if len(df) == 0:
        print('[!] No images found. Check data directory and folder structure.')
        return df

    print(f'\nHyperKvasir loaded: {len(df)} images across {df["source_class"].nunique()} classes')
    print('\nRisk-tier distribution:')
    for label in range(NUM_CLASSES):
        subset = df[df['label'] == label]
        print(f'    Class {label} | {LABEL_NAMES[label]:<20}: {len(subset):>5}')
    return df

```

DATA_ROOT = 'data/HyperKvasir'

df = load_hyperkvasir(DATA_ROOT, max_normal=3000)

```

if len(df) > 0:
    train_df, temp = train_test_split(df, test_size=0.3, stratify=df['label'])
    val_df, test_df = train_test_split(temp, test_size=0.5, stratify=temp['label'])
    print(f'\nSplit: train={len(train_df)} val={len(val_df)} test={len(test_df)}')

```

[INFO] Normal class capped: 5252 → 3000 images

HyperKvasir loaded: 7633 images across 21 classes

Risk-tier distribution:

Class 0 Normal	:	3000	(39.3%)
Class 1 Inflammatory	:	645	(8.5%)
Class 2 Pre_malignant	:	1836	(24.1%)
Class 3 High_Risk	:	2152	(28.2%)

Split: train=5343 val=1145 test=1145

In [4]: **def** visualize_dataset_samples(data_dir, label_map, label_names, n_per_class=4,
 """Sample images per risk class with solid colored sidebar labels (GridSpec)

```

from collections import defaultdict
import matplotlib.gridspec as _gs

if not os.path.exists(data_dir):
    print(f'(!) Directory not found: {data_dir}'); return

class_images = defaultdict(list)
for root, _, files in os.walk(data_dir):
    folder = os.path.basename(root)
    label = label_map.get(folder)
    if label is None:
        continue
    class_images[label].extend(
        os.path.join(root, f) for f in files
        if f.lower().endswith(('.jpg', '.jpeg', '.png')))

n_classes = len(label_names)
colors = ['#2ECC71', '#3498DB', '#F39C12', '#E74C3C']

fig = plt.figure(figsize=(n_per_class * 2.8 + 1.4, n_classes * 2.8))
fig.suptitle(f'{title} - Sample Images per Risk Class', fontsize=13, fontw

gs = fig.add_gridspec(
    n_classes, n_per_class + 1,
    width_ratios=[0.38] + [1] * n_per_class,
    hspace=0.06, wspace=0.04,
    top=0.96, bottom=0.02, left=0.02, right=0.98
)

for cls in range(n_classes):
    # Solid colored sidebar with white class name
    ax_lbl = fig.add_subplot(gs[cls, 0])
    ax_lbl.set_facecolor(colors[cls])
    ax_lbl.text(0.5, 0.5, label_names[cls],
               ha='center', va='center', fontsize=10, fontweight='bold',
               color='white', rotation=90, transform=ax_lbl.transAxes)
    ax_lbl.set_xticks([]); ax_lbl.set_yticks([])
    for spine in ax_lbl.spines.values(): spine.set_visible(False)

    imgs = class_images.get(cls, [])
    sample = random.sample(imgs, min(n_per_class, len(imgs)))

    for col in range(n_per_class):
        ax = fig.add_subplot(gs[cls, col + 1])
        if col < len(sample):
            try: ax.imshow(Image.open(sample[col]).convert('RGB'))
            except: ax.set_facecolor('#cccccc')
        else:
            ax.set_facecolor('#cccccc')
        ax.axis('off')
        for spine in ax.spines.values():
            spine.set_visible(True)
            spine.set_edgecolor(colors[cls])

```

```

        spine.set_linewidth(1.5)

    out = os.path.join(PROJECT_DIR, f'samples_{title.replace(" ", "_").lower()}')
    plt.savefig(out, dpi=120, bbox_inches='tight', facecolor='white')
    plt.show()
    print(f'Saved: {out}')

print('visualize_dataset_samples() ready.')

```

visualize_dataset_samples() ready.

```

In [5]: def plot_class_distribution(data_dir, label_map, label_names, title='Class Dis
        from collections import defaultdict
        counts = defaultdict(int)

        if not os.path.exists(data_dir):
            print(f'[] Directory not found: {data_dir}'); return

        # os.walk handles both flat (Kvasir-v2) and nested (HyperKvasir) structure
        for root, dirs, files in os.walk(data_dir):
            folder_name = os.path.basename(root)
            label = label_map.get(folder_name)
            if label is None:
                continue
            counts[label] += sum(1 for f in files
                                if f.lower().endswith(('.jpg', '.jpeg', '.png')))

        total = sum(counts.values())
        if total == 0:
            print(f'[] No images found in {data_dir}'); return

        colors = ['#2ECC71', '#3498DB', '#F39C12', '#E74C3C']
        fig, axes = plt.subplots(1, 2, figsize=(13, 4))

        bars = axes[0].bar([label_names[i] for i in range(len(label_names))],
                           [counts[i] for i in range(len(label_names))],
                           color=colors, alpha=0.85, edgecolor='white')
        for bar, i in zip(bars, range(len(label_names))):
            axes[0].text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 5,
                          f'{counts[i]}\n({100*counts[i]/total:.1f}%',
                          ha='center', va='bottom', fontsize=9, fontweight='bold')
        axes[0].set_ylabel('Image Count', title=title)
        axes[0].grid(axis='y', alpha=0.3)

        axes[1].pie([counts[i] for i in range(len(label_names))],
                     labels=label_names, colors=colors, autopct='%1.1f%%',
                     startangle=90, pctdistance=0.8)
        axes[1].set_title(f'{title} - Pie')

        plt.tight_layout()
        plt.savefig(os.path.join(PROJECT_DIR, f'dist_{title.replace(" ", "_").lower()}'))
        plt.show()

    print('plot_class_distribution() ready.')

```

```
plot_class_distribution() ready.
```

[illegible]

[!] Run the KVASIR_V2_MAP cell in Section 8 first, then re-run this cell.

[illegible]

[!] Run the KVASIR_V2_MAP cell in Section 8 first, then re-run this cell.

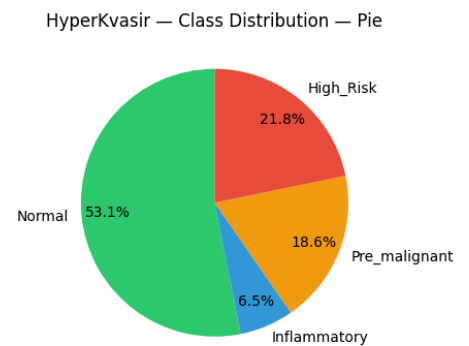
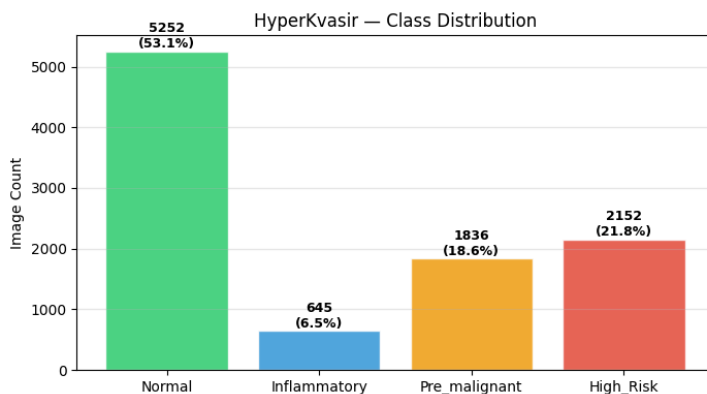
[illegible]

HyperKvasir — Sample Images per Risk Class



Saved: D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\samples_hyperkvasir.png

```
In [9]: # HyperKvasir class distribution (run after extraction)
plot_class_distribution('data/HyperKvasir/labelled-images', HYPERKVASIR_MAP, LA
title='HyperKvasir — Class Distribution')
```



```

In [10]: def plot_imbalance_handling():
    fig, axes = plt.subplots(1, 3, figsize=(16, 5))
    colors = ['#2ECC71', '#3498DB', '#F39C12', '#E74C3C']
    labels = ['Normal', 'Inflammatory', 'Pre-malignant', 'High-Risk']
    x = np.arange(len(labels))
    w = 0.35

    # — Panel 1: Raw vs Capped —————
    raw = [5252, 645, 1836, 2152]
    capped = [3000, 645, 1836, 2152]
    axes[0].bar(x - w/2, raw, w, label='Raw (before cap)', color=colors,
    axes[0].bar(x + w/2, capped, w, label='After cap (training)', color=colors
    for i, val in enumerate(capped):
        axes[0].text(x[i] + w/2, val + 30, str(val), ha='center', fontsize=8,
    axes[0].axhline(3000, color='red', linestyle='--', linewidth=1.2, label='N
    axes[0].set(xticks=x, xticklabels=labels, ylabel='Image Count',
        title='① Normal Cap (max_normal=3000)\nReduces dominant clas
    axes[0].tick_params(axis='x', rotation=12); axes[0].legend(fontsize=8); ax

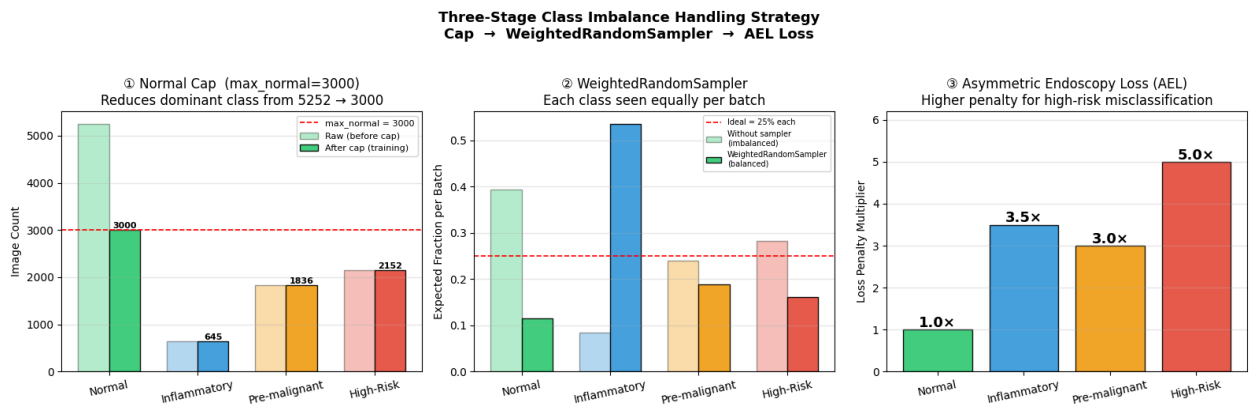
    # — Panel 2: WeightedRandomSampler – expected probability per batch —————
    counts = np.array([3000, 645, 1836, 2152], dtype=float)
    proportional = counts / counts.sum()
    inv_w = 1.0 / counts
    balanced = inv_w / inv_w.sum()
    axes[1].bar(x - w/2, proportional, w, label='Without sampler\n(imbalanced)
    axes[1].bar(x + w/2, balanced, w, label='WeightedRandomSampler\n(balan
    axes[1].axhline(0.25, color='red', linestyle='--', linewidth=1.2, label='I
    axes[1].set(xticks=x, xticklabels=labels, ylabel='Expected Fraction per Ba
        title='② WeightedRandomSampler\nEach class seen equally per b
    axes[1].tick_params(axis='x', rotation=12); axes[1].legend(fontsize=7); ax

    # — Panel 3: AEL class weights —————
    ael = [1.0, 3.5, 3.0, 5.0]
    bars = axes[2].bar(labels, ael, color=colors, alpha=0.9, edgecolor='black'
    for bar, val in zip(bars, ael):
        axes[2].text(bar.get_x() + bar.get_width()/2, val + 0.06,
            f'{val}x', ha='center', fontsize=13, fontweight='bold')
    axes[2].set(ylabel='Loss Penalty Multiplier', ylim=(0, 6.2),
        title='③ Asymmetric Endoscopy Loss (AEL)\nHigher penalty for l
    axes[2].tick_params(axis='x', rotation=12); axes[2].grid(axis='y', alpha=0

    plt.suptitle('Three-Stage Class Imbalance Handling Strategy\n'
        'Cap → WeightedRandomSampler → AEL Loss',
        fontsize=13, fontweight='bold', y=1.03)
    plt.tight_layout()
    plt.savefig(os.path.join(PROJECT_DIR, 'imbalance_handling.png'), dpi=150,
    plt.show()
    print('Saved → imbalance_handling.png')

plot_imbalance_handling()

```



Saved → imbalance_handling.png

3. Data Augmentation & DataLoader

Preprocessing pipeline:

Image → CLAHE Enhancement → Resize → Augmentation → Normalize → Tensor

CLAHE (Contrast Limited Adaptive Histogram Equalization) enhances mucosal texture, vascular patterns, and pit structures critical for risk-level discrimination.

```
In [11]: from PIL import ImageFile
ImageFile.LOAD_TRUNCATED_IMAGES = True # Tolerate partially-extracted images

class CLAHETransform:
    """CLAHE on the L-channel of LAB space to enhance mucosal texture."""
    def __init__(self, clip_limit=CLAHE_CLIP_LIMIT, tile_grid_size=CLAHE_TILE_SIZE):
        self.clip_limit = clip_limit
        self.tile_grid_size = tile_grid_size

    def __call__(self, img):
        try:
            img_np = np.array(img.convert('RGB'))
            clahe = cv2.createCLAHE(clipLimit=self.clip_limit,
                                    tileGridSize=self.tile_grid_size)
            lab = cv2.cvtColor(img_np, cv2.COLOR_RGB2LAB)
            lab[..., 0] = clahe.apply(lab[..., 0])
            img_np = cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)
            return Image.fromarray(img_np)
        except Exception as e:
            warnings.warn(f'CLAHETransform failed ({e}); returning original image')
            return img

DATASET_MEAN = [0.485, 0.456, 0.406]
DATASET_STD = [0.229, 0.224, 0.225]
```

```

train_transforms = transforms.Compose([
    CLAHETransform(),
    transforms.Resize((IMG_SIZE + 32, IMG_SIZE + 32)),
    transforms.RandomCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.RandomVerticalFlip(p=0.2),
    transforms.RandomRotation(degrees=15),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.1),
    transforms.ToTensor(),
    transforms.Normalize(mean=DATASET_MEAN, std=DATASET_STD),
])

val_transforms = transforms.Compose([
    CLAHETransform(),
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(),
    transforms.Normalize(mean=DATASET_MEAN, std=DATASET_STD),
])

_BLANK = Image.new('RGB', (IMG_SIZE, IMG_SIZE), (0, 0, 0))

class EndoscopyDataset(Dataset):
    def __init__(self, df, transform=None):
        self.df = df.reset_index(drop=True)
        self.transform = transform
        missing = set(p for p in self.df['image_path'] if not os.path.exists(p))
        if missing:
            warnings.warn(f'{len(missing)} image paths not found – those rows')
            self.df = self.df[~self.df['image_path'].isin(missing)].reset_index()

    def __len__(self):
        return len(self.df)

    def __getitem__(self, idx):
        row = self.df.iloc[idx]
        try:
            img = Image.open(row['image_path']).convert('RGB')
        except Exception as e:
            warnings.warn(f"Bad image skipped: {e}")
            img = _BLANK.copy()
        if self.transform:
            img = self.transform(img)
        return img, torch.tensor(row['label'], dtype=torch.long)

def make_loaders(train_df, val_df, test_df):
    counts = train_df['label'].value_counts().sort_index().values
    weights = 1.0 / counts
    sw = torch.tensor(weights[train_df['label'].values], dtype=torch.float)
    sampler = WeightedRandomSampler(
        sw, num_samples=len(sw), replacement=True,
        generator=torch.Generator().manual_seed(SEED)
    )

```

```

)
pin = DEVICE.type == 'cuda'
train_ldr = DataLoader(EndoscopyDataset(train_df, train_transforms),
                        BATCH_SIZE, sampler=sampler, num_workers=0, pin_memory=pin)
val_ldr = DataLoader(EndoscopyDataset(val_df, val_transforms),
                     BATCH_SIZE, shuffle=False, num_workers=0, pin_memory=pin)
test_ldr = DataLoader(EndoscopyDataset(test_df, val_transforms),
                      BATCH_SIZE, shuffle=False, num_workers=0, pin_memory=pin)
return train_ldr, val_ldr, test_ldr

if len(df) > 0:
    train_ldr, val_ldr, test_ldr = make_loaders(train_df, val_df, test_df)
    print(f'Dataloaders ready: train={len(train_ldr)} batches val={len(val_ldr)} batches test={len(test_ldr)} batches')
    print('Pipeline: CLAHE -> Resize -> Augment -> Normalize')
    print('Truncated images: PIL loads with gray fill (LOAD_TRUNCATED_IMAGES=True)')

```

Dataloaders ready: train=167 batches val=36 test=36
 Pipeline: CLAHE -> Resize -> Augment -> Normalize
 Truncated images: PIL loads with gray fill (LOAD_TRUNCATED_IMAGES=True)

4. Model Architecture — DenseNet-121 · EfficientNet-B0 · DeiT-Tiny

Lightweight CNN vs. Transformer benchmark — all models <8M parameters, suitable for real-time inference on endoscopy hardware.

Model	Type	Pre-training	Parameters	Dropout
DenseNet-121	CNN	ImageNet-1k	7.0M	0.5
EfficientNet-B0	CNN	ImageNet-1k	5.3M	0.3
DeiT-Tiny	Transformer	ImageNet-1k	5.9M	0.1

```

In [12]: import timm

def build_densenet121(num_classes=NUM_CLASSES):
    m = models.densenet121(weights='IMAGENET1K_V1')
    m.classifier = nn.Sequential(
        nn.Dropout(DROPOUT_RATES['densenet121']),
        nn.Linear(m.classifier.in_features, num_classes)
    )
    return m

def build_efficientnet_b0(num_classes=NUM_CLASSES):
    m = timm.create_model('efficientnet_b0', pretrained=True, num_classes=0)
    m.classifier = nn.Sequential(
        nn.Dropout(DROPOUT_RATES['efficientnet_b0']),
        nn.Linear(m.num_features, 256),
        nn.ReLU(),
        nn.Dropout(0.2),
    )

```



```

        self.ce = nn.CrossEntropyLoss(weight=w)

    def forward(self, logits, targets):
        return self.ce(logits, targets)

print('Asymmetric Endoscopy Loss (AEL):')
for i, (name, risk, w) in enumerate(zip(LABEL_NAMES, RISK_LABELS, AEL_WEIGHTS)):
    print(f'  Class {i} | {name:<20} weight={w} → {risk}')

```

Asymmetric Endoscopy Loss (AEL):

Class 0	Normal	weight=1.0	→ Routine Surveillance
Class 1	Inflammatory	weight=3.5	→ Medical Management
Class 2	Pre_malignant	weight=3.0	→ Biopsy + Surveillance
Class 3	High_Risk	weight=5.0	→ Immediate Intervention

6. Training with Transfer Learning

In [14]: **import** json

```

def train_epoch(model, loader, optimizer, scheduler, criterion, device):
    model.train()
    total_loss, preds_all, trues_all = 0.0, [], []
    for imgs, labels in tqdm(loader, desc='  batches', leave=True):
        imgs, labels = imgs.to(device), labels.to(device)
        optimizer.zero_grad()
        out = model(imgs)
        loss = criterion(out, labels)
        loss.backward()
        nn.utils.clip_grad_norm_(model.parameters(), 1.0)
        optimizer.step()
        total_loss += loss.item()
        preds_all.extend(out.detach().argmax(1).cpu().numpy())
        trues_all.extend(labels.cpu().numpy())
    scheduler.step()
    return (total_loss / len(loader),
            f1_score(trues_all, preds_all, average='macro', zero_division=0))

@torch.no_grad()
def evaluate(model, loader, criterion, device):
    model.eval()
    total_loss, preds_all, trues_all = 0.0, [], []
    for imgs, labels in loader:
        imgs, labels = imgs.to(device), labels.to(device)
        out = model(imgs)
        total_loss += criterion(out, labels).item()
        preds_all.extend(out.argmax(1).cpu().numpy())
        trues_all.extend(labels.cpu().numpy())
    assert len(preds_all) == len(trues_all)
    return (f1_score(trues_all, preds_all, average='macro', zero_division=0),
            total_loss / len(loader), preds_all, trues_all)

```

```

def _save_history(history, hist_path):
    tmp = hist_path + '.tmp'
    with open(tmp, 'w') as f:
        json.dump(history, f)
    os.replace(tmp, hist_path)

def _save_checkpoint(model_state, optimizer_state, scheduler_state,
                    epoch, best_f1, history, ckpt_dir):
    """Save full training state – enables resume after crash."""
    resume_path = os.path.join(ckpt_dir, 'resume.pt')
    tmp = resume_path + '.tmp'
    torch.save({
        'epoch': epoch,
        'best_f1': best_f1,
        'model_state': model_state,
        'optimizer_state': optimizer_state,
        'scheduler_state': scheduler_state,
    }, tmp)
    os.replace(tmp, resume_path)

def _save_best(model_state, ckpt_path):
    tmp = ckpt_path + '.tmp'
    torch.save(model_state, tmp)
    os.replace(tmp, ckpt_path)

def run_training(model, train_ldr, val_ldr, model_name,
                epochs=25, lr=2e-4, ckpt_dir=CKPT_DIR):
    ckpt_path = os.path.join(ckpt_dir, model_name, 'best.pt')
    hist_path = os.path.join(ckpt_dir, model_name, 'history.json')
    resume_path = os.path.join(ckpt_dir, model_name, 'resume.pt')
    os.makedirs(os.path.dirname(ckpt_path), exist_ok=True)

    model = model.to(DEVICE)
    criterion = AsymmetricEndoscopyLoss()
    optimizer = AdamW(model.parameters(), lr=lr, weight_decay=1e-4)
    scheduler = CosineAnnealingLR(optimizer, T_max=epochs, eta_min=1e-6)

    start_epoch = 1
    best_f1 = 0.0
    history = {'train_loss': [], 'val_loss': [], 'train_f1': [], 'val_f1': []}

    # — Resume from interrupted run —
    if os.path.exists(resume_path):
        state = torch.load(resume_path, map_location=DEVICE, weights_only=False)
        completed = state['epoch']
        if completed >= epochs:
            print(f'[DONE] {model_name} already completed {completed}/{epochs}')
            model.load_state_dict(torch.load(ckpt_path, map_location=DEVICE, w

```

```

        with open(hist_path) as f:
            history = json.load(f)
        return model, history
    # Partial run – resume from next epoch
    model.load_state_dict(state['model_state'])
    optimizer.load_state_dict(state['optimizer_state'])
    scheduler.load_state_dict(state['scheduler_state'])
    best_f1 = state['best_f1']
    start_epoch = completed + 1
    with open(hist_path) as f:
        history = json.load(f)
    print(f'[RESUME] {model_name} – continuing from epoch {start_epoch}/{epochs} (best val F1 so far: {best_f1:.4f})')

elif os.path.exists(ckpt_path):
    # Old-style checkpoint (no resume.pt) – start fresh
    completed_epochs = len(json.load(open(hist_path)).get('train_loss', []))
    if os.path.exists(hist_path) else 0
    if completed_epochs >= epochs:
        print(f'[DONE] {model_name} – loading finished checkpoint.')
        model.load_state_dict(torch.load(ckpt_path, map_location=DEVICE, weights_only=True))
        with open(hist_path) as f:
            history = json.load(f)
        return model, history
    print(f'[WARN] Checkpoint found but no resume state. Re-training from scratch')

else:
    print(f'\n{"="*60}')
    print(f'  Training : {model_name}')
    print(f'  Device   : {DEVICE} | Epochs: {epochs} | LR: {lr}')
    print(f'  Loss     : AsymmetricEndoscopyLoss weights={AEL_WEIGHTS}')
    print(f'  Saves    : history + resume state every epoch | best.pt on disk')
    print(f'{"="*60}')

for epoch in range(start_epoch, epochs + 1):
    print(f'\nEpoch {epoch}/{epochs}')
    tl, tf = train_epoch(model, train_loader, optimizer, scheduler, criterion)
    vf, vl, _, _ = evaluate(model, val_loader, criterion, DEVICE)

    history['train_loss'].append(tl)
    history['val_loss'].append(vl)
    history['train_f1'].append(tf)
    history['val_f1'].append(vf)

    # Save full resume state + history every epoch
    _save_checkpoint(copy.deepcopy(model.state_dict()),
                    optimizer.state_dict(), scheduler.state_dict(),
                    epoch, best_f1, history,
                    os.path.dirname(ckpt_path))
    _save_history(history, hist_path)

    marker = ''
    if vf > best_f1:

```

```

        best_f1 = vf
        _save_best(copy.deepcopy(model.state_dict()), ckpt_path)
        marker = ' *** NEW BEST – model saved ***'

    print(f' loss={tl:.4f} train_F1={tf:.4f} '
          f'val_loss={vl:.4f} val_F1={vf:.4f}{marker}')

    # Reload best weights
    model.load_state_dict(torch.load(ckpt_path, map_location=DEVICE, weights_c
    print(f'\n Done. Best val F1 = {best_f1:.4f}')
    print(f' Checkpoint → {ckpt_path}')
    print(f' History → {hist_path} ({len(history["train_loss"])} epochs)'
    return model, history

def plot_training_curves(histories: dict):
    fig, axes = plt.subplots(1, 2, figsize=(13, 5))
    colors = ['#2980B9', '#E74C3C', '#2ECC71', '#F39C12']
    for (name, h), color in zip(histories.items(), colors):
        if not h:
            continue
        axes[0].plot(h['train_loss'], '--', color=color, alpha=0.6, label=f'{name} train_loss')
        axes[0].plot(h['val_loss'], '-', color=color, label=f'{name} val_loss')
        axes[1].plot(h['train_f1'], '--', color=color, alpha=0.6)
        axes[1].plot(h['val_f1'], '-', color=color, label=f'{name} val_f1')
    for ax, title, ylabel in zip(axes,
                                ['AEL Loss Curves', 'Macro F1 Curves'],
                                ['Loss', 'Macro F1']):
        ax.set(xlabel='Epoch', ylabel=ylabel, title=title)
        ax.legend(fontsize=8)
        ax.grid(alpha=0.3)
    plt.suptitle('Training History – CNN vs Transformer (HyperKvasir)',
                 fontsize=13, fontweight='bold')
    plt.tight_layout()
    plt.savefig(os.path.join(PROJECT_DIR, 'training_curves.png'), dpi=150, bbox_inches='tight')
    plt.show()

print('Training functions defined.')
print('Saves: history.json + resume.pt every epoch | best.pt on new best val F1')
print('Resume: re-run any training cell after crash – continues from last completed epoch')

```

Training functions defined.

Saves: history.json + resume.pt every epoch | best.pt on new best val F1

Resume: re-run any training cell after crash – continues from last completed epoch

```

In [15]: # — SETUP: initialise result dicts (run once before training) —————
if 'train_ldr' not in globals():
    raise RuntimeError('Run Sections 2 & 3 first to load data and build DataLoader')

if 'trained_models' not in globals():
    trained_models = {}
if 'histories' not in globals():
    histories = {}

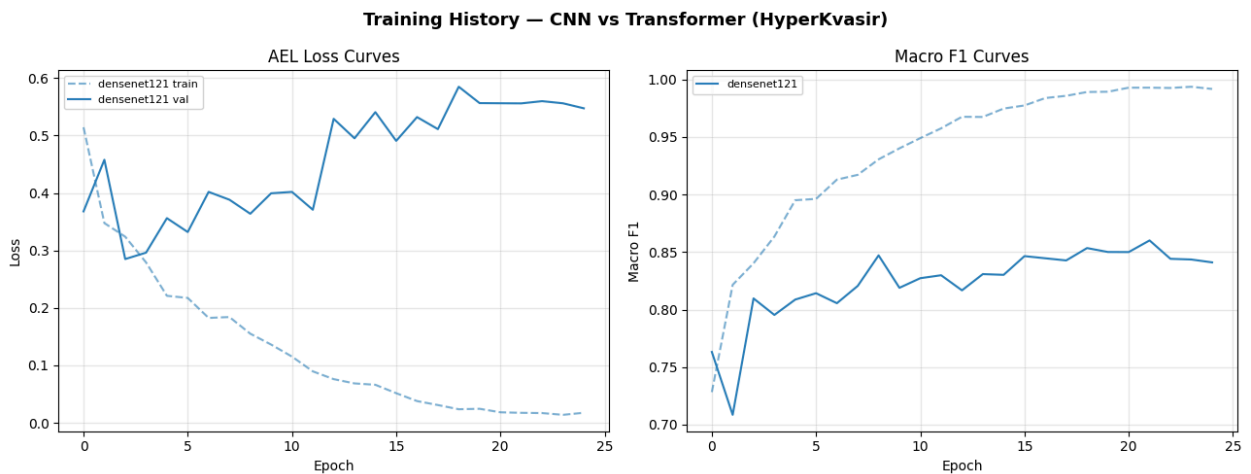
```

```
print('Result dicts ready. Run each model cell below in order.')
print('Each cell saves a checkpoint – safe to resume after interruption.')
```

Result dicts ready. Run each model cell below in order.
Each cell saves a checkpoint – safe to resume after interruption.

```
In [16]: # — MODEL 1 / 3 : DenseNet-121 —————
# Checkpoint: checkpoints/densenet121/best.pt
# If interrupted, re-run this cell – it will load the saved checkpoint automat
trained_models['densenet121'], histories['densenet121'] = \
    run_training(build_densenet121(), train_ldr, val_ldr, 'densenet121')
plot_training_curves(histories) # auto-save training_curves.png
print('DenseNet-121 done.')
```

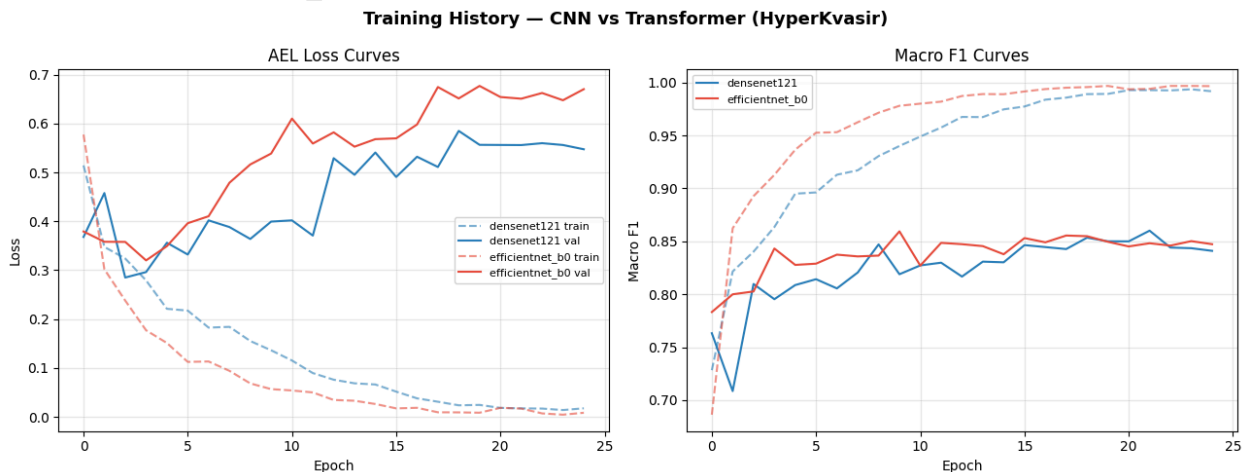
[DONE] densenet121 already completed 25/25 epochs.



DenseNet-121 done.

```
In [17]: # — MODEL 2 / 3 : EfficientNet-B0 —————
trained_models['efficientnet_b0'], histories['efficientnet_b0'] = \
    run_training(build_efficientnet_b0(), train_ldr, val_ldr, 'efficientnet_b0')
plot_training_curves(histories) # auto-save training_curves.png
print('EfficientNet-B0 done.')
```

[DONE] efficientnet_b0 already completed 25/25 epochs.

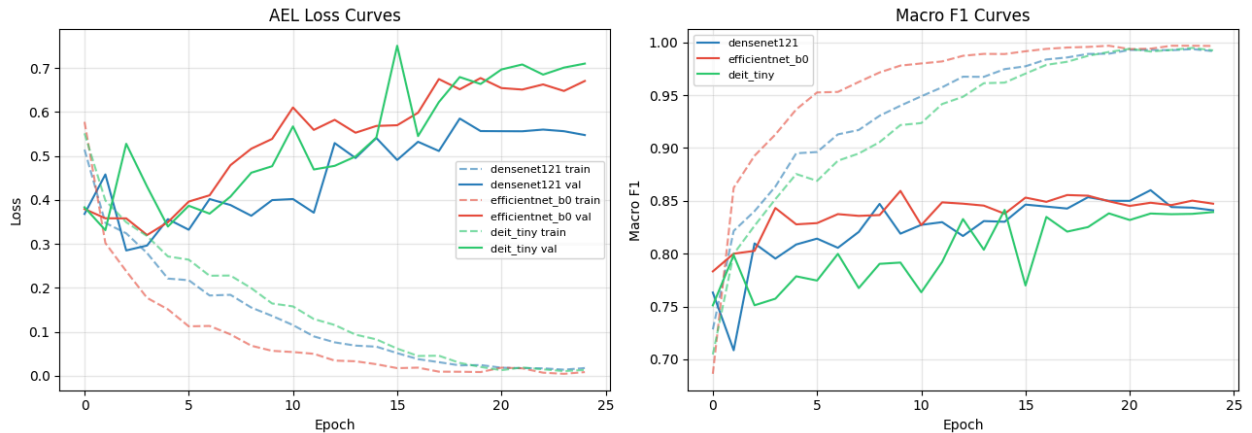


EfficientNet-B0 done.

```
In [18]: # — MODEL 3 / 3 : DeiT-Tiny —————
trained_models['deit_tiny'], histories['deit_tiny'] = \
    run_training(build_deit_tiny(), train_ldr, val_ldr, 'deit_tiny')
plot_training_curves(histories) # auto-save training_curves.png
print('DeiT-Tiny done.')
```

[DONE] deit_tiny already completed 25/25 epochs.

Training History — CNN vs Transformer (HyperKvasir)

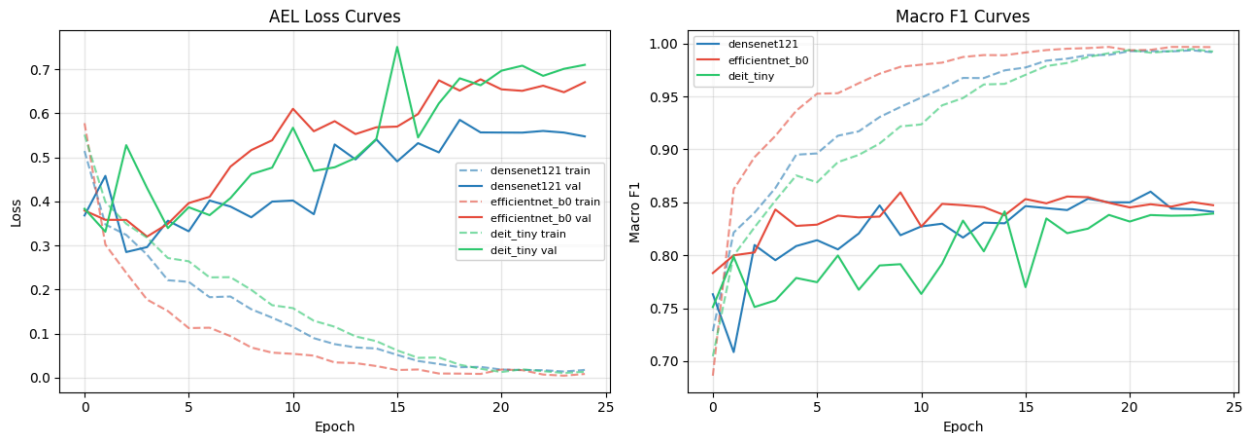


DeiT-Tiny done.

```
In [19]: # — PLOT TRAINING CURVES (run after all 4 models complete) —————
print(f'Models trained so far: {list(trained_models.keys())}')
if histories:
    plot_training_curves(histories)
```

Models trained so far: ['densenet121', 'efficientnet_b0', 'deit_tiny']

Training History — CNN vs Transformer (HyperKvasir)



7. AEL Ablation Study

Compare Asymmetric Endoscopy Loss vs Cross-Entropy vs Focal Loss on DenseNet-121.

```
In [20]: class FocalLoss(nn.Module):
def __init__(self, gamma=2.0):
    super().__init__()
```

```

        self.gamma = gamma

    def forward(self, logits, targets):
        ce = F.cross_entropy(logits, targets, reduction='none')
        pt = torch.exp(-ce)
        return ((1 - pt) ** self.gamma * ce).mean()

def run_ablation(train_ldr, val_ldr, test_ldr, epochs=25,
                 progress_path=None):
    """Train 3 loss variants on DenseNet-121.
    Saves one JSON entry per variant so a crash only costs the current variant
    """
    import json as _j, os as _o

    # Load any already-finished variants
    done = {}
    if progress_path and _o.path.exists(progress_path):
        done = _j.load(open(progress_path))
        print(f'[RESUME] Quick ablation: {list(done.keys())} already done.')

    loss_fns = {
        'Cross-Entropy': nn.CrossEntropyLoss(),
        'Focal (γ=2)': FocalLoss(gamma=2.0),
        'AEL (Ours)': AsymmetricEndoscopyLoss(),
    }
    results = dict(done)

    for name, loss_fn in loss_fns.items():
        if name in done:
            print(f'[SKIP] {name} already done.')
            continue
        print(f'\n>>> Ablation: {name}')
        model = build_densenet121().to(DEVICE)
        opt = AdamW(model.parameters(), lr=2e-4, weight_decay=1e-4)
        sch = CosineAnnealingLR(opt, T_max=epochs, eta_min=1e-6)
        for epoch in range(1, epochs + 1):
            model.train()
            running_loss = 0.0
            batch_bar = tqdm(train_ldr, desc=f'{name} Ep{epoch}/{epochs}',
                             unit='batch', leave=False)
            for imgs, labels in batch_bar:
                imgs, labels = imgs.to(DEVICE), labels.to(DEVICE)
                opt.zero_grad()
                loss = loss_fn(model(imgs), labels)
                loss.backward()
                nn.utils.clip_grad_norm_(model.parameters(), 1.0)
                opt.step()
                running_loss += loss.item()
                batch_bar.set_postfix(loss=f'{loss.item():.4f}')
            sch.step()
        print(f'Ep {epoch}/{epochs} - avg loss: {running_loss / len(train_ldr)}')

```

```

model.eval()
preds_all, trues_all = [], []
with torch.no_grad():
    for imgs, labels in tqdm(test_ldr, desc=f' {name} eval', leave=False):
        preds_all.extend(model(imgs.to(DEVICE)).argmax(1).cpu().numpy())
        trues_all.extend(labels.numpy())
macro_f1 = f1_score(trues_all, preds_all, average='macro', zero_division=0)
high_risk_recall = recall_score(
    [1 if t == 3 else 0 for t in trues_all],
    [1 if p == 3 else 0 for p in preds_all], zero_division=0)
results[name] = {'macro_f1': macro_f1, 'high_risk_recall': high_risk_recall}
print(f' Done - Macro F1={macro_f1:.4f} High-Risk Recall={high_risk_recall:.4f}')

# Atomic save after each variant so a crash costs at most one variant
if progress_path:
    _tmp = progress_path + '.tmp'
    with open(_tmp, 'w') as _f:
        _j.dump(results, _f, indent=2)
    _o.replace(_tmp, progress_path)

print('\nAblation Summary:')
print(f'{"Loss":<20} {"Macro F1":>10} {"High-Risk Recall":>18}')
print('-' * 50)
for k, v in results.items():
    print(f'{"k":<20} {"v["macro_f1"]":>10.4f} {"v["high_risk_recall"]":>18.4f}')
return results

print('Ablation ready. Call: ablation_results = run_ablation(train_ldr, val_ldr, test_ldr)')

```

Ablation ready. Call: ablation_results = run_ablation(train_ldr, val_ldr, test_ldr)

```

In [21]: # — RUN ABLATION STUDY (≈ 75 min on M2 – 3 × 25 epochs) —————
# Requires: train_ldr, val_ldr, test_ldr (run Sections 2-3 first)
import json as _json, os as _os
import matplotlib.pyplot as plt
import numpy as np

_abl_prog = _os.path.join(CKPT_DIR, 'ablation_quick_progress.json')
ablation_results = run_ablation(train_ldr, val_ldr, test_ldr, epochs=25,
                                progress_path=_abl_prog)

# — Save final JSON for reproducibility —————
_abl_path = _os.path.join(CKPT_DIR, 'ablation_results.json')
with open(_abl_path, 'w') as _f:
    _json.dump(ablation_results, _f, indent=2)
print(f'Saved → {_abl_path}')

# — Bar chart —————
_names = list(ablation_results.keys())
_f1s = [ablation_results[k]['macro_f1'] for k in _names]
_hr_rec = [ablation_results[k]['high_risk_recall'] for k in _names]

```

```

_x = np.arange(len(_names))
_w = 0.35
_fig, _ax = plt.subplots(figsize=(8, 5))
_b1 = _ax.bar(_x - _w/2, _fls, _w, label='Macro F1', color='steelbl
_b2 = _ax.bar(_x + _w/2, _hr_rec, _w, label='High-Risk Recall', color='tomato'
_ax.set_xticks(_x)
_ax.set_xticklabels(_names, fontsize=12)
_ax.set_ylim(0, 1.10)
_ax.set_ylabel('Score', fontsize=12)
_ax.set_title('AEL Ablation Study – DenseNet-121 (25 epochs)', fontsize=13)
_ax.legend(fontsize=11)
_ax.bar_label(_b1, fmt='%.3f', padding=3, fontsize=9)
_ax.bar_label(_b2, fmt='%.3f', padding=3, fontsize=9)
plt.tight_layout()
_png = os.path.join(PROJECT_DIR, 'ablation_study.png')
plt.savefig(_png, dpi=150, bbox_inches='tight')
plt.show()
print(f'Saved → {_png}')

# — Print LaTeX rows to paste into paper.tex —————
print('\n=== LaTeX rows for tab:ablation ===')
_label_map = {
    'AEL (Ours)': r'AEL  $[1.0, 3.5, 3.0, 5.0]$  (Ours)',
    'Cross-Entropy': r'Cross-Entropy (uniform)',
    'Focal ( $\gamma=2$ ):': r'Focal Loss ( $\gamma=2$ )',
}
for k, v in ablation_results.items():
    label = _label_map.get(k, k)
    print(f'{label} & {v["macro_f1"]:.4f} & {v["high_risk_recall"]:.4f}\\\\')

```

[RESUME] Quick ablation: ['Cross-Entropy', 'Focal ($\gamma=2$)', 'AEL (Ours)'] already done.

[SKIP] Cross-Entropy already done.

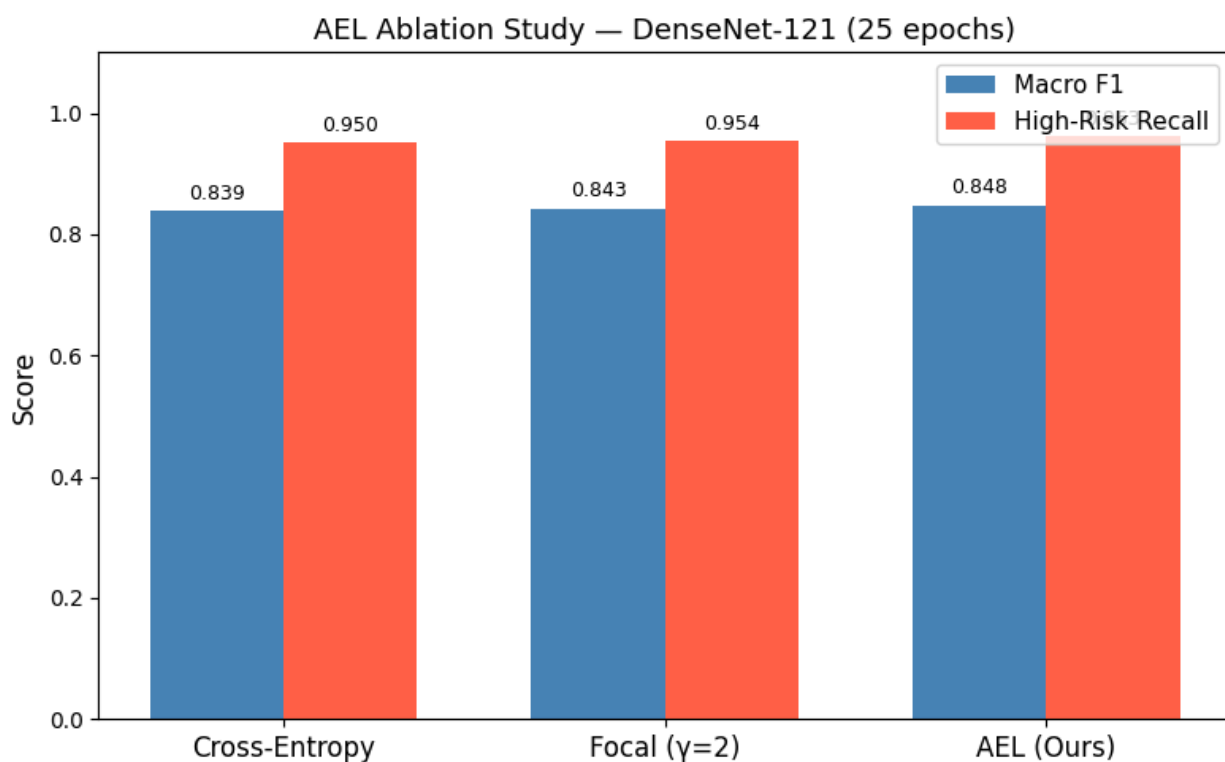
[SKIP] Focal ($\gamma=2$) already done.

[SKIP] AEL (Ours) already done.

Ablation Summary:

Loss	Macro F1	High-Risk Recall
Cross-Entropy	0.8393	0.9505
Focal ($\gamma=2$)	0.8428	0.9536
AEL (Ours)	0.8477	0.9628

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```
=== LaTeX rows for tab:ablation ===
Cross-Entropy (uniform) & 0.8393 & 0.9505\\
Focal Loss ( $\gamma=2$ ) & 0.8428 & 0.9536\\
AEL  $[1.0, 3.5, 3.0, 5.0]$  (Ours) & 0.8477 & 0.9628\\
```

8. Cross-Dataset Evaluation — Kvasir-v2

Train on HyperKvasir, evaluate zero-shot on Kvasir-v2. Kvasir-v2 folder structure:
data/Kvasir-v2//

```
In [22]: KVASIR_V2_MAP = {
    'esophagitis': 1, # Inflammatory
    'ulcerative-colitis': 2, # Pre-malignant (Kvasir-v2 has mixed severity)
    'polyps': 2, # Pre-malignant
    'barretts': 2, # Pre-malignant
    'normal-cecum': 0,
    'normal-pylorus': 0,
    'normal-z-line': 0,
    'dyed-lifted-polyps': 3, # High-Risk
    'dyed-resection-margins': 3,
}

def load_kvasir_v2(data_dir: str) -> pd.DataFrame:
    rows = []
    if not os.path.exists(data_dir):
```

```

    print(f'(!) Kvasir-v2 not found at {data_dir}')
    print('    Download from: https://datasets.simula.no/kvasir/')
    return pd.DataFrame()
for class_folder in os.listdir(data_dir):
    class_dir = os.path.join(data_dir, class_folder)
    if not os.path.isdir(class_dir): continue
    label = KVASIR_V2_MAP.get(class_folder)
    if label is None: continue
    for f in os.listdir(class_dir):
        if not f.lower().endswith(('.jpg', '.jpeg', '.png')): continue
        rows.append({'image_path': os.path.join(class_dir, f),
                    'label': label, 'label_name': LABEL_NAMES[label],
                    'source_class': class_folder})
df = pd.DataFrame(rows)
if len(df):
    print(f'Kvasir-v2: {len(df)} images')
    print(df['label_name'].value_counts().to_string())
return df

@torch.no_grad()
def evaluate_detailed(model, loader, device, dataset_name=''):
    model.eval()
    preds_all, trues_all, probs_all = [], [], []
    with torch.no_grad():
        for imgs, labels in tqdm(loader, desc=f'Evaluating {dataset_name}', leave=True):
            out = model(imgs.to(device))
            probs_all.extend(F.softmax(out, dim=1).cpu().numpy())
            preds_all.extend(out.argmax(1).cpu().numpy())
            trues_all.extend(labels.numpy())
    macro_f1 = f1_score(trues_all, preds_all, average='macro', zero_division=0)
    per_class = f1_score(trues_all, preds_all, average=None, zero_division=0)
    high_risk_recall = recall_score(
        [1 if t == 3 else 0 for t in trues_all],
        [1 if p == 3 else 0 for p in preds_all], zero_division=0)
    print(f'\n{"="*60}')
    print(f'Dataset: {dataset_name} | Macro F1: {macro_f1:.4f} | High-Risk Recall: {high_risk_recall:.4f}')
    print(classification_report(trues_all, preds_all, target_names=LABEL_NAMES))
    return {'macro_f1': macro_f1, 'per_class': per_class,
            'high_risk_recall': high_risk_recall,
            'preds': preds_all, 'trues': trues_all,
            'probs': np.array(probs_all)}

def plot_confusion_matrix(results, model_name):
    cm = confusion_matrix(results['trues'], results['preds'])
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5))
    short = ['Normal', 'Inflam.', 'Pre-mal.', 'High-Risk']
    for ax, data, fmt, title in zip([ax1, ax2], [cm, cm.astype(float) / cm.sum(axis=1, keepdims=True)],
                                    ['d', '.1%'], ['Count', 'Normalized']):
        sns.heatmap(data, annot=True, fmt=fmt, cmap='Blues', ax=ax, title=title)

```

```

        xticklabels=short, yticklabels=short)
    ax.set(xlabel='Predicted', ylabel='True',
           title=f'{model_name} Confusion Matrix ({title})')
    plt.tight_layout()
    plt.savefig(os.path.join(PROJECT_DIR, f'confusion_matrix_{model_name}.png'))
    plt.show()

print('Cross-dataset evaluation ready.')

```

Cross-dataset evaluation ready.

```

In [23]: # — RUN EVALUATION (test set + Kvasir-v2) —————
criterion = AsymmetricEndoscopyLoss()
results = {}

for name, model in trained_models.items():
    print(f'\n--- {name} ---')
    results[name] = evaluate_detailed(model, test_ldr, DEVICE,
                                     dataset_name=f'{name} (HyperKvasir test)')
    plot_confusion_matrix(results[name], name)

# Cross-dataset on Kvasir-v2
kv2_df = load_kvasir_v2('data/Kvasir-v2')
ext_results = {}
if len(kv2_df) > 0:
    kv2_ldr = DataLoader(EndoscopyDataset(kv2_df, val_transforms),
                        BATCH_SIZE, shuffle=False, num_workers=0)
    ext_results['Kvasir-v2'] = {}
    for name, model in trained_models.items():
        print(f'\n--- {name} on Kvasir-v2 ---')
        ext_results['Kvasir-v2'][name] = evaluate_detailed(
            model, kv2_ldr, DEVICE,
            dataset_name=f'{name} (Kvasir-v2)')

print('Evaluation complete. Run Section 9 for ROC curves, Section 14 for paper)

# — Save main HyperKvasir results to CSV —————
import pandas as _pd_eval
_main_rows = []
for _name, _res in results.items():
    _main_rows.append({
        'model': _name,
        'macro_f1': round(_res['macro_f1'], 6),
        'high_risk_recall': round(_res['high_risk_recall'], 6),
        'normal_f1': round(float(_res['per_class'][0]), 6),
        'inflammatory_f1': round(float(_res['per_class'][1]), 6),
        'prealignant_f1': round(float(_res['per_class'][2]), 6),
        'highrisk_f1': round(float(_res['per_class'][3]), 6),
    })
_main_csv = os.path.join(PROJECT_DIR, 'main_results.csv')
_pd_eval.DataFrame(_main_rows).to_csv(_main_csv, index=False)
print(f'Saved \u2192 {_main_csv}')

```

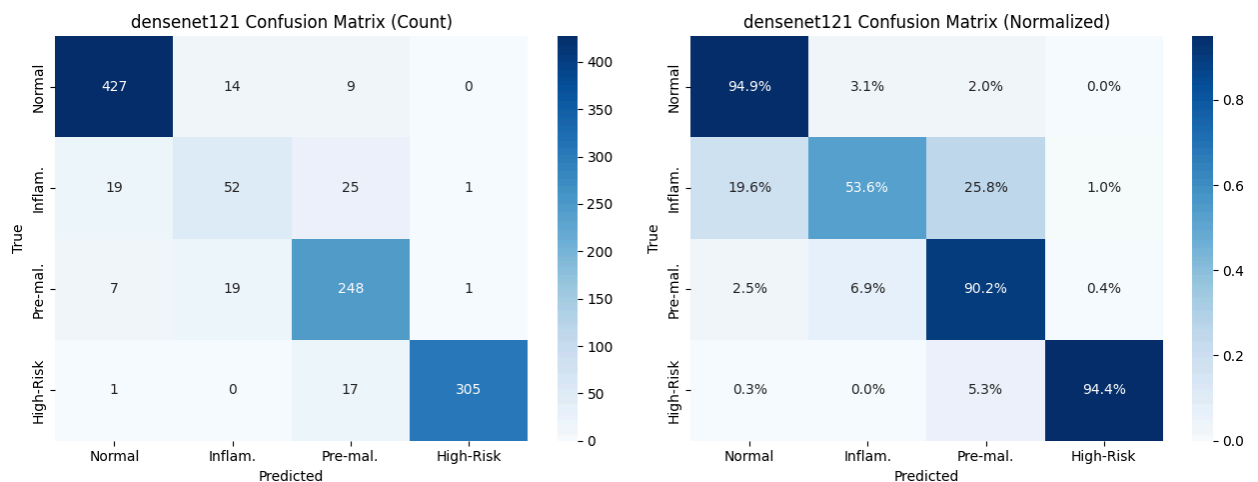
```
# — Save Kvasir-v2 cross-dataset results to CSV —————
if ext_results:
    _kv2_rows = []
    for _name in results:
        _kv2 = ext_results.get('Kvasir-v2', {}).get(_name, {})
        if _kv2:
            _kv2_rows.append({
                'model': _name,
                'hk_macro_f1': round(results[_name]['macro_f1'], 6),
                'kv2_macro_f1': round(_kv2['macro_f1'], 6),
                'f1_drop': round(results[_name]['macro_f1'] - _kv2['macro_f1'], 6),
                'kv2_hr_recall': round(_kv2['high_risk_recall'], 6),
            })
    _kv2_csv = os.path.join(PROJECT_DIR, 'kv2_results.csv')
    _pd_eval.DataFrame(_kv2_rows).to_csv(_kv2_csv, index=False)
    print(f'Saved \u2192 {_kv2_csv}')
```

--- densenet121 ---

Evaluating densenet121 (HyperKvasir test): 100%|██████████| 36/36 [02:11<00:00, 3.66s/it]

Dataset: densenet121 (HyperKvasir test) | Macro F1: 0.8371 | High-Risk Recall: 0.9443

	precision	recall	f1-score	support
Normal	0.94	0.95	0.94	450
Inflammatory	0.61	0.54	0.57	97
Pre_malignant	0.83	0.90	0.86	275
High_Risk	0.99	0.94	0.97	323
accuracy			0.90	1145
macro avg	0.84	0.83	0.84	1145
weighted avg	0.90	0.90	0.90	1145

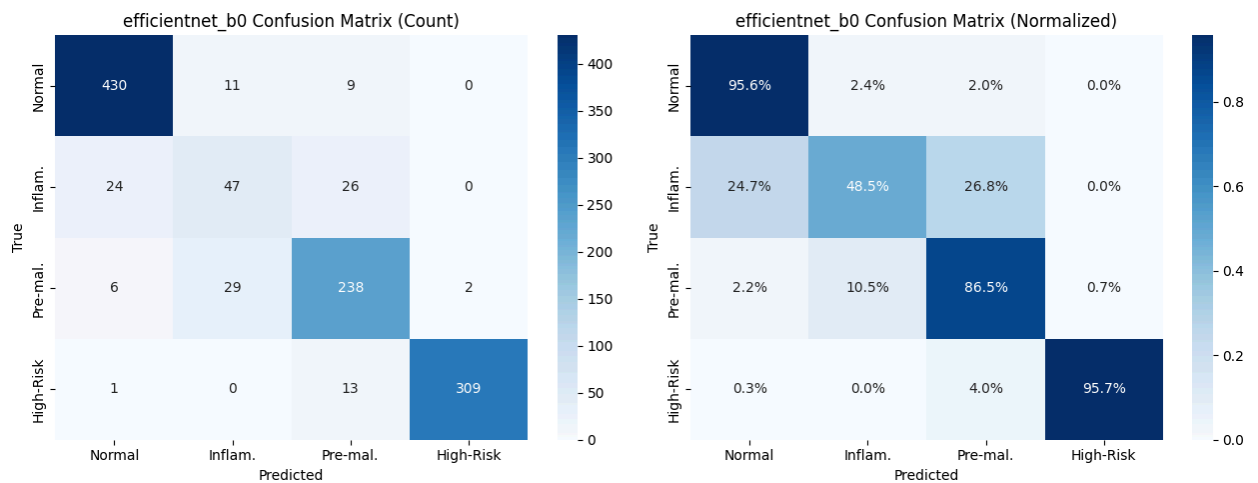


--- efficientnet_b0 ---

Evaluating efficientnet_b0 (HyperKvasir test): 100%|██████████| 36/36 [00:49<00:00, 1.38s/it]

Dataset: efficientnet_b0 (HyperKvasir test) | Macro F1: 0.8195 | High-Risk Recall: 0.9567

	precision	recall	f1-score	support
Normal	0.93	0.96	0.94	450
Inflammatory	0.54	0.48	0.51	97
Pre_malignant	0.83	0.87	0.85	275
High_Risk	0.99	0.96	0.97	323
accuracy			0.89	1145
macro avg	0.82	0.82	0.82	1145
weighted avg	0.89	0.89	0.89	1145

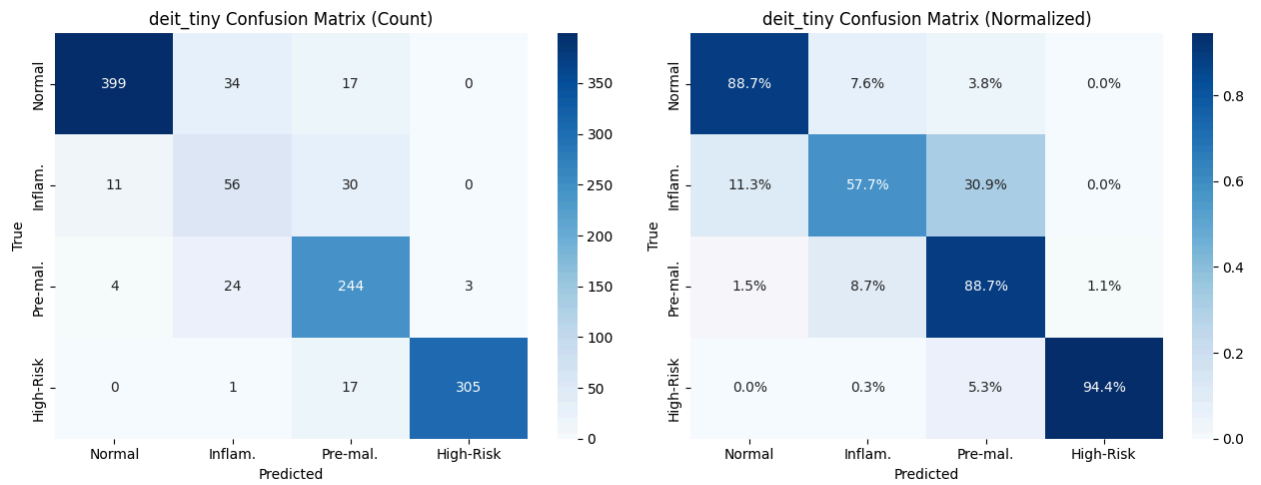


--- deit_tiny ---

Evaluating deit_tiny (HyperKvasir test): 100% | 36/36 [00:39<00:00, 1.11s/it]

Dataset: deit_tiny (HyperKvasir test) | Macro F1: 0.8139 | High-Risk Recall: 0.9443

	precision	recall	f1-score	support
Normal	0.96	0.89	0.92	450
Inflammatory	0.49	0.58	0.53	97
Pre_malignant	0.79	0.89	0.84	275
High_Risk	0.99	0.94	0.97	323
accuracy			0.88	1145
macro avg	0.81	0.82	0.81	1145
weighted avg	0.89	0.88	0.88	1145



Kvasir-v2: 8000 images

label_name

Normal 3000

High_Risk 2000

Pre_malignant 2000

Inflammatory 1000

--- densenet121 on Kvasir-v2 ---

Evaluating densenet121 (Kvasir-v2): 100% | ██████████ | 250/250 [08:10<00:00, 1.96s/it]

=====

Dataset: densenet121 (Kvasir-v2) | Macro F1: 0.7307 | High-Risk Recall: 0.8725

	precision	recall	f1-score	support
Normal	0.96	0.69	0.80	3000
Inflammatory	0.75	0.37	0.50	1000
Pre_malignant	0.54	0.97	0.70	2000
High_Risk	0.99	0.87	0.93	2000
accuracy			0.77	8000
macro avg	0.81	0.73	0.73	8000
weighted avg	0.84	0.77	0.77	8000

--- efficientnet_b0 on Kvasir-v2 ---

Evaluating efficientnet_b0 (Kvasir-v2): 100% | ██████████ | 250/250 [07:09<00:00, 1.72s/it]

```
=====
Dataset: efficientnet_b0 (Kvasir-v2) | Macro F1: 0.7756 | High-Risk Recall: 0.9965
```

	precision	recall	f1-score	support
Normal	0.94	0.87	0.90	3000
Inflammatory	0.56	0.43	0.48	1000
Pre_malignant	0.69	0.81	0.74	2000
High_Risk	0.95	1.00	0.98	2000
accuracy			0.83	8000
macro avg	0.78	0.78	0.78	8000
weighted avg	0.83	0.83	0.83	8000

```
--- deit_tiny on Kvasir-v2 ---
```

```
Evaluating deit_tiny (Kvasir-v2): 100%|██████████| 250/250 [08:32<00:00, 2.05s/it]
```

```
=====
Dataset: deit_tiny (Kvasir-v2) | Macro F1: 0.7555 | High-Risk Recall: 0.9875
```

	precision	recall	f1-score	support
Normal	0.97	0.64	0.77	3000
Inflammatory	0.57	0.48	0.52	1000
Pre_malignant	0.61	0.96	0.74	2000
High_Risk	0.98	0.99	0.99	2000
accuracy			0.79	8000
macro avg	0.78	0.77	0.76	8000
weighted avg	0.83	0.79	0.79	8000

Evaluation complete. Run Section 9 for ROC curves, Section 14 for paper tables.
 Saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\main_results.csv
 Saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\kv2_results.csv

9. ROC-AUC Curves — Per Class (One-vs-Rest)

```
In [24]: def plot_roc_curves(results_dict: dict):
    n_cls = NUM_CLASSES
    colors = ['#2ECC71', '#3498DB', '#F39C12', '#E74C3C']
    styles = ['-', '--', ':', '-.']
    fig, axes = plt.subplots(1, n_cls, figsize=(5 * n_cls, 5))

    for c, (cls_name, risk, ax) in enumerate(zip(LABEL_NAMES, RISK_LABELS, axes)):
        for (model_name, results), ls in zip(results_dict.items(), styles):
            trues_bin = [1 if t == c else 0 for t in results['trues']]
            probs_cls = np.array(results['probs'])[ :, c]
            fpr, tpr, _ = roc_curve(trues_bin, probs_cls)
            roc_auc = auc(fpr, tpr)
```

```

        ax.plot(fpr, tpr, linestyle=ls, linewidth=2,
                label=f'{model_name} (AUC={roc_auc:.3f})')
    ax.plot([0, 1], [0, 1], 'k--', linewidth=1, alpha=0.5)
    ax.set(xlabel='FPR', ylabel='TPR',
           title=f'{cls_name}\n({risk})')
    ax.legend(fontsize=7); ax.grid(alpha=0.3)
    if c == 3:
        ax.text(0.3, 0.05, 'HIGH PRIORITY\nImmediate Intervention',
                fontsize=8, color='red', alpha=0.8,
                bbox=dict(boxstyle='round', facecolor='wheat', alpha=0.3))

plt.suptitle('Per-Class ROC Curves – GI Lesion Risk Stratification (One-vs
            fontsize=13, fontweight='bold')
plt.tight_layout()
plt.savefig(os.path.join(PROJECT_DIR, 'roc_curves.png'), dpi=150, bbox_inches=
plt.show()

def print_auc_table(results_dict: dict):
    print(f'\n{"Model":<22}', end='')
    for c in LABEL_NAMES: print(f'{c:>16}', end='')
    print(f'{"Macro AUC":>12}')
    print('-' * 90)
    for model_name, results in results_dict.items():
        trues_bin = label_binarize(results['trues'], classes=list(range(NUM_CLASSES)))
        aucs = []
        print(f'{model_name:<22}', end='')
        for c in range(NUM_CLASSES):
            a = roc_auc_score(trues_bin[:, c], np.array(results['probs'])[c])
            aucs.append(a)
            print(f'{a:>16.4f}', end='')
        print(f'{np.mean(aucs):>12.4f}')

print('ROC-AUC analysis ready.')

# — Run ROC analysis (requires Section 8 evaluation to have run first) —
if 'results' in globals() and results:
    plot_roc_curves(results)
    print_auc_table(results)
else:
    print('[!] Run Section 8 evaluation first.')

# — Save AUC results to CSV —
if 'results' in globals() and results:
    import pandas as _pd_auc
    from sklearn.preprocessing import label_binarize
    from sklearn.metrics import roc_auc_score
    _auc_rows = []
    for _name, _res in results.items():
        _tb = label_binarize(_res['trues'], classes=list(range(NUM_CLASSES)))
        _aucs = [roc_auc_score(_tb[:, c], np.array(_res['probs'])[c]) for c in range(NUM_CLASSES)]

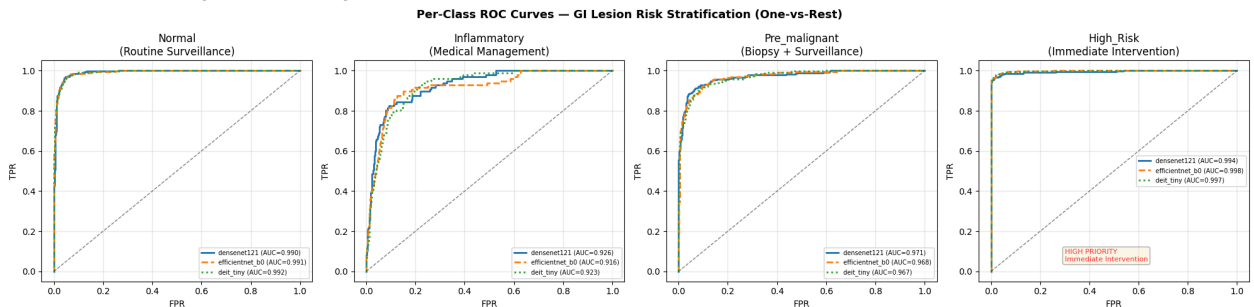
```

```

    _auc_rows.append({
        'model': _name,
        'auc_normal': round(_aucs[0], 6),
        'auc_inflammatory': round(_aucs[1], 6),
        'auc_premalignant': round(_aucs[2], 6),
        'auc_highrisk': round(_aucs[3], 6),
        'macro_auc': round(float(np.mean(_aucs)), 6),
    })
    _auc_csv = os.path.join(PROJECT_DIR, 'auc_results.csv')
    _pd_auc.DataFrame(_auc_rows).to_csv(_auc_csv, index=False)
    print(f'AUC results saved \u2192 {_auc_csv}')

```

ROC-AUC analysis ready.



Model		Normal	Inflammatory	Pre_malignant	Hi
gh_Risk	Macro AUC				
densenet121		0.9901	0.9265	0.9707	
0.9941	0.9704				
efficientnet_b0		0.9910	0.9155	0.9682	
0.9978	0.9681				
deit_tiny		0.9916	0.9231	0.9673	
0.9974	0.9699				

AUC results saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\auc_results.csv

10. GradCAM Explainability

Localizes mucosal irregularities, pit patterns, and vascular changes that drive each risk-tier classification decision.

```

In [25]: class GradCAM:
    def __init__(self, model, target_layer):
        self.model = model
        self.grads = None
        self.acts = None
        target_layer.register_forward_hook(
            lambda m, i, o: setattr(self, 'acts', o.detach()))
        # register_full_backward_hook is the non-deprecated API (PyTorch ≥ 2.0)
        target_layer.register_full_backward_hook(
            lambda m, gi, go: setattr(self, 'grads', go[0].detach()))

    def generate(self, img_tensor, class_idx=None):

```

```

        self.model.eval()
        img = img_tensor.unsqueeze(0).to(DEVICE)
        out = self.model(img)
        if class_idx is None:
            class_idx = out.argmax().item()
        self.model.zero_grad()
        out[0, class_idx].backward()

        if self.grads.dim() == 4:
            # CNN: (B, C, H, W)
            weights = self.grads.mean(dim=[2, 3], keepdim=True)
            cam = F.relu((weights * self.acts).sum(1, keepdim=True))
            cam = cam.squeeze().cpu().numpy()
        else:
            # Transformer: (B, N, C) – reshape token sequence to 2-D spatial g
            weights = self.grads.mean(dim=1) # (E
            cam = F.relu((weights.unsqueeze(1) * self.acts).sum(-1)) # (E
            cam = cam.squeeze().cpu().numpy()
            n = int(cam.shape[0] ** 0.5)
            cam = cam[:n * n].reshape(n, n)

        cam = (cam - cam.min()) / (cam.max() - cam.min() + 1e-8)
        cam = cv2.resize(cam, (IMG_SIZE, IMG_SIZE))
        return cam, class_idx

def get_target_layer(model, model_name):
    if model_name == 'densenet121':
        return model.features.denseblock4.denselayer16.conv2
    elif model_name == 'efficientnet_b0':
        return model.conv_head # last 1x1 conv before global pool
    elif model_name == 'deit_tiny':
        return model.blocks[-1].norm1 # final transformer block layer norm
    raise ValueError(f'Unknown model: {model_name}')

def visualize_gradcam(model, model_name, image_paths, true_labels=None):
    target_layer = get_target_layer(model, model_name)
    gcam = GradCAM(model, target_layer)
    inv = transforms.Normalize(
        mean=[-m / s for m, s in zip(DATASET_MEAN, DATASET_STD)],
        std=[1 / s for s in DATASET_STD])

    n = len(image_paths)
    colors = ['#2ECC71', '#3498DB', '#F39C12', '#E74C3C']
    fig, axes = plt.subplots(2, n, figsize=(4 * n, 9))
    axes = axes.reshape(2, n) # always 2-D, regardless of n

    for i, path in enumerate(image_paths):
        img_t = val_transforms(Image.open(path).convert('RGB'))
        cam, pred = gcam.generate(img_t)
        img_np = np.array(inv(img_t).permute(1, 2, 0).clamp(0, 1))
        heatmap = cv2.cvtColor(

```

```

        cv2.applyColorMap(np.uint8(255 * cam), cv2.COLORMAP_JET),
        cv2.COLOR_BGR2RGB) / 255.0
    overlay = np.clip(0.6 * img_np + 0.4 * heatmap, 0, 1)

    gt = LABEL_NAMES[true_labels[i]] if true_labels else ''
    axes[0, i].imshow(img_np); axes[0, i].set_title(f'GT: {gt}', fontsize=12)
    axes[1, i].imshow(overlay); axes[1, i].set_title(
        f'{LABEL_NAMES[pred]}\n{RISK_LABELS[pred]}',
        fontsize=9, color=colors[pred])
    for ax in [axes[0, i], axes[1, i]]:
        ax.axis('off')

    plt.suptitle(f'GradCAM - {model_name}',
                 fontsize=12, fontweight='bold')
    plt.tight_layout()
    save_path = os.path.join(
        os.path.dirname(os.path.abspath('GastroEndoscopy_Risk_Stratification.ipynb')),
        f'gradcam_{model_name}.png')
    plt.savefig(save_path, dpi=150, bbox_inches='tight')
    print(f'Saved: {save_path}')
    plt.show()

print('GradCAM ready.')

```

GradCAM ready.

In [26]: **import** os

```

# — RUN GRADCAM — one image per risk class, all trained models —————
SAVE_DIR = os.path.dirname(os.path.abspath('GastroEndoscopy_Risk_Stratification.ipynb'))

# Pick one representative test image per class
sample_paths, sample_labels = [], []
for label in range(NUM_CLASSES):
    rows = test_df[test_df['label'] == label]
    if len(rows) == 0:
        continue
    row = rows.iloc[0]
    sample_paths.append(row['image_path'])
    sample_labels.append(label)

print(f'Generating GradCAM for {len(sample_paths)} classes...')

import matplotlib
matplotlib.use('Agg') # save to file without needing a display
import matplotlib.pyplot as plt

for name, model in trained_models.items():
    out_path = os.path.join(SAVE_DIR, f'gradcam_{name}.png')
    visualize_gradcam(model, name, sample_paths, true_labels=sample_labels)
    # rename if saved with relative path
    if os.path.exists(f'gradcam_{name}.png') and not os.path.exists(out_path):
        os.rename(f'gradcam_{name}.png', out_path)
    print(f'  Saved → {out_path}')

```

```
print('Done. PNG files saved in project folder.')
```

Generating GradCAM for 4 classes...

Saved: D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\gradcam_densenet121.png

Saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\gradcam_densenet121.png

Saved: D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\gradcam_efficientnet_b0.png

Saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\gradcam_efficientnet_b0.png

Saved: D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\gradcam_deit_tiny.png

Saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\gradcam_deit_tiny.png

Done. PNG files saved in project folder.

11. Monte Carlo Dropout Uncertainty

Flags borderline lesions for mandatory endoscopist review.

```
In [27]: def mc_predict(model, img_tensor, n_samples=30):
    """Monte Carlo Dropout inference. Runs n_samples stochastic forward passes
    model.eval()
    # Enable only Dropout layers for MC sampling
    for m in model.modules():
        if isinstance(m, nn.Dropout):
            m.train()

    img = img_tensor.unsqueeze(0).to(DEVICE)
    probs = []
    with torch.no_grad():
        for _ in range(n_samples):
            probs.append(F.softmax(model(img), dim=1).cpu().numpy())

    # Restore full eval mode
    model.eval()

    probs = np.stack(probs).squeeze(1)          # (n_samples, NUM_CLASSES)
    mean_probs = probs.mean(axis=0)
    pred = int(mean_probs.argmax())
    confidence = float(mean_probs.max())
    entropy = float(-np.sum(mean_probs * np.log(mean_probs + 1e-8)))
    flag = confidence < 0.75 or pred >= 2        # Pre-malignant or High-Risk

    return {
        'prediction': pred,
        'label': LABEL_NAMES[pred],
        'risk': RISK_LABELS[pred],
        'confidence': confidence,
        'uncertainty': entropy,
```

```

        'flag':          flag,
        'mean_probs':    mean_probs,
    }

print('MC Dropout ready.')
print('Flag rule: confidence < 0.75 OR predicted class ≥ Pre-malignant')

```

MC Dropout ready.

Flag rule: confidence < 0.75 OR predicted class ≥ Pre-malignant

12. Endoscopist Workload Simulation

Quantifies AI burden reduction while maintaining zero missed High-Risk lesions.

```

In [28]: def endoscopist_simulation(model, test_df, confidence_threshold=0.75):
    auto_cleared, flagged = 0, 0
    missed_high_risk = 0
    flag_breakdown = {'low_confidence': 0, 'pre_malignant': 0, 'high_risk': 0}

    # Run mc_predict once per image and store results
    mc_results = []
    for _, row in tqdm(test_df.iterrows(), total=len(test_df), desc='Simulation'):
        img = val_transforms(Image.open(row['image_path']).convert('RGB'))
        res = mc_predict(model, img)
        mc_results.append((res, row['label']))

    # Apply the caller-supplied threshold (not mc_predict's hardcoded 0.75)
    for res, true_label in mc_results:
        auto = res['confidence'] >= confidence_threshold and res['prediction'] == 0
        if not auto:
            flagged += 1
            if res['confidence'] < confidence_threshold: flag_breakdown['low_confidence'] += 1
            if res['prediction'] == 2: flag_breakdown['pre_malignant'] += 1
            if res['prediction'] == 3: flag_breakdown['high_risk'] += 1
        else:
            auto_cleared += 1
            if true_label == 3:
                missed_high_risk += 1

    total = len(test_df)
    print(f'\nEndoscopist Workload Simulation (threshold={confidence_threshold})')
    print(f'  Total patients: {total}')
    print(f'  AI auto-cleared: {auto_cleared} ({100*auto_cleared/total:.1f}%)')
    print(f'  Flagged for endoscopist: {flagged} ({100*flagged/total:.1f}%)')
    print(f'  Workload reduction: {100*auto_cleared/total:.1f}%')
    print(f'  Missed High-Risk lesions: {missed_high_risk} (target: 0)')
    print(f'  Flag breakdown: {flag_breakdown}')

    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 5))
    ax1.pie([auto_cleared, flagged],
            labels=[f'AI auto-cleared\n({100*auto_cleared/total:.1f}%)',

```

```

        f'Endoscopist review\n({100*flagged/total:.1f}%)',
        colors=['#2ECC71', '#E74C3C'], autopct='%1.1f%%', startangle=90)
ax1.set_title('Endoscopy Screening Workflow\nAI Workload Distribution')

# Threshold sweep over precomputed mc_results – no redundant forward passes
thresholds = [0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90]
reductions = []
for thresh in thresholds:
    cleared = sum(1 for res, _ in mc_results
                  if res['confidence'] >= thresh and res['prediction'] < 2)
    reductions.append(100 * cleared / total)
ax2.plot(thresholds, reductions, 'b-o', linewidth=2)
ax2.axvline(confidence_threshold, color='red', linestyle='--',
            label=f'Selected threshold={confidence_threshold}')
ax2.set(xlabel='Confidence Threshold', ylabel='Workload Reduction (%)',
        title='Threshold vs. Endoscopist Burden Reduction')
ax2.legend(); ax2.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(PROJECT_DIR, 'workload_simulation.png'), dpi=150,
            plt.show())
return auto_cleared, flagged, missed_high_risk

print('Simulation ready.')

```

Simulation ready.

```

In [29]: # — RUN ENDOSCOPIST WORKLOAD SIMULATION —————
if 'results' not in globals() or not results:
    raise RuntimeError('Run Section 8 evaluation first – results dict is missing')
best_model_name = max(trained_models.keys(),
                      key=lambda n: results[n]['macro_f1'] if n in results else 0)
best_model = trained_models[best_model_name]
print(f'Running simulation with best model: {best_model_name}')
auto_cleared, flagged, missed = endoscopist_simulation(
    best_model, test_df, confidence_threshold=0.75)
print(f'PNG saved → {os.path.join(PROJECT_DIR, "workload_simulation.png")}')

# — Save workload results to JSON and CSV —————
import json as _json_wl, pandas as _pd_wl
_total = len(test_df)
_workload_data = {
    'model': best_model_name,
    'total': _total,
    'auto_cleared': int(auto_cleared),
    'flagged': int(flagged),
    'missed_high_risk': int(missed),
    'pct_auto_cleared': round(100 * auto_cleared / _total, 1),
    'pct_flagged': round(100 * flagged / _total, 1),
    'threshold': 0.75,
    'mc_passes': 30,
}
_wl_json = os.path.join(PROJECT_DIR, 'workload_results.json')
with open(_wl_json, 'w') as _f:

```

```

    _json_wl.dump(_workload_data, _f, indent=2)
_wl_csv = os.path.join(PROJECT_DIR, 'workload_results.csv')
_pd_wl.DataFrame(_workload_data).to_csv(_wl_csv, index=False)
print(f'Workload results saved \u2192 {_wl_json}')
print(f'Workload results saved \u2192 {_wl_csv}')
print(f'  Auto-cleared: {_workload_data["auto_cleared"]} ({_workload_data["pct_"]}')
print(f'  Flagged:      {_workload_data["flagged"]} ({_workload_data["pct_flag"]}')
print(f'  Missed HR:    {_workload_data["missed_high_risk"]}'))

```

Running simulation with best model: densenet121

Simulating: 100%|██████████| 1145/1145 [25:57<00:00, 1.36s/it]

Endoscopist Workload Simulation (threshold=0.75)

```

Total patients:          1145
AI auto-cleared:         514 (44.9%)
Flagged for endoscopist: 631 (55.1%)
Workload reduction:      44.9%
Missed High-Risk lesions: 0 (target: 0)
Flag breakdown: {'low_confidence': 40, 'pre_malignant': 299, 'high_risk': 30
7}

```

PNG saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\workload_simulation.png

Workload results saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\workload_results.json

Workload results saved → D:\Research paper code run\gas\GastroEndoscopy-Risk-Stratification\workload_results.csv

```

Auto-cleared: 514 (44.9%)
Flagged:      631 (55.1%)
Missed HR:    0

```

13. Multi-Seed AEL Sensitivity — High-Risk Weight Sweep (Table 11)

Sweeps the High-Risk AEL penalty weight over {3.0, 4.0, 5.0, 6.0, 7.0} with N=3 seeds each. Run this section FIRST — the best weight is automatically passed to Section 14 (Ablation).

```

In [ ]: import random, numpy as np, torch, torch.nn as nn, json, os
        from torch.utils.data import DataLoader
        from sklearn.metrics import f1_score

        for _v in ['train_df', 'val_df', 'test_df']:
            if _v not in globals():
                raise RuntimeError(f'Run Sections 2 & 3 first ({_v} missing).')

SENS_SEEDS      = [42, 0, 123]
SENS_EPOCHS     = 25
SENS_LR         = 1e-4
SENS_DROPOUT    = 0.5
SENS_HR_WEIGHTS = [3.0, 4.0, 5.0, 6.0, 7.0]
SENS_BASE_W     = [1.0, 3.5, 3.0]
PROGRESS_FILE   = os.path.join(PROJECT_DIR, 'checkpoints', 'sensitivity_notebo

```

```

os.makedirs(os.path.dirname(PROGRESS_FILE), exist_ok=True)

if os.path.exists(PROGRESS_FILE):
    with open(PROGRESS_FILE) as _f:
        _saved = json.load(_f)
        print(f'Resuming - {sum(len(v) for v in _saved.values())} seed-runs already saved')
    else:
        _saved = {}

_sens_train_tf = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.RandomHorizontalFlip(), transforms.RandomVerticalFlip(),
    transforms.ColorJitter(brightness=0.2, contrast=0.2),
    transforms.ToTensor(), transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.229, 0.229])
])
_sens_val_tf = transforms.Compose([
    transforms.Resize((IMG_SIZE, IMG_SIZE)),
    transforms.ToTensor(), transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.229, 0.229])
])

_counts = train_df['label'].value_counts().sort_index().values
_w_per = 1.0 / _counts
_samples = torch.tensor([_w_per[l] for l in train_df['label'].values], dtype=torch.float)
_sampler = WeightedRandomSampler(_samples, len(_samples), replacement=True,
                                generator=torch.Generator().manual_seed(42))
_test_ldr = DataLoader(EndoscopyDataset(test_df, _sens_val_tf), BATCH_SIZE, shuffle=True)

def _build_sens_model():
    import torchvision.models as _m
    m = _m.densenet121(weights=None)
    m.classifier = nn.Sequential(nn.Dropout(SENS_DROPOUT), nn.Linear(m.classifier.fc.in_features, SENS_CLASSES))
    return m.to(DEVICE)

def _train_sens(ael_weights, hr_w, seed):
    random.seed(seed); np.random.seed(seed); torch.manual_seed(seed)
    model = _build_sens_model()
    criterion = nn.CrossEntropyLoss(weight=torch.tensor(ael_weights, dtype=torch.float))
    optimizer = torch.optim.Adam(model.parameters(), lr=SENS_LR, weight_decay=SENS_WEIGHT_DECAY)
    scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=SENS_EPOCHS)
    _train_ldr = DataLoader(EndoscopyDataset(train_df, _sens_train_tf), BATCH_SIZE, shuffle=True)
    _val_ldr = DataLoader(EndoscopyDataset(val_df, _sens_val_tf), BATCH_SIZE, shuffle=False)
    best_f1 = 0.0; best_state = {k: v.clone() for k, v in model.state_dict().items()}
    for ep in range(1, SENS_EPOCHS + 1):
        model.train()
        for imgs, labels in _train_ldr:
            imgs, labels = imgs.to(DEVICE), labels.to(DEVICE)
            optimizer.zero_grad(); criterion(model(imgs), labels).backward();
            scheduler.step()
        model.eval(); preds, trues = [], []
        with torch.no_grad():
            for imgs, labels in _val_ldr:
                preds.extend(model(imgs.to(DEVICE)).argmax(1).cpu().numpy());
        vf1 = f1_score(trues, preds, average='macro', zero_division=0)

```

```

        print(f'    [w={hr_w} seed={seed}] ep {ep}/{SENS_EPOCHS} val_f1={vf1:.4f}'
              f'if vf1 > best_f1: best_f1 = vf1; best_state = {k: v.clone() for k, v in'
              f'model.load_state_dict(best_state)}; return model')

@torch.no_grad()
def _eval_sens(model):
    model.eval(); preds, trues = [], []
    for imgs, labels in _test_loader:
        preds.extend(model(imgs.to(DEVICE)).argmax(1).cpu().numpy()); trues.extend(
            labels.to(DEVICE).cpu().numpy())
    preds, trues = np.array(preds), np.array(trues)
    macro_f1 = float(f1_score(trues, preds, average='macro', zero_division=0))
    mask = trues == 3
    hr_rec = float((preds[mask] == 3).sum() / mask.sum()) if mask.sum() > 0 else 0
    return macro_f1, hr_rec

sens_progress = {str(w): _saved.get(str(w), {}) for w in SENS_HR_WEIGHTS}

for w in SENS_HR_WEIGHTS:
    ael = SENS_BASE_W + [w]
    wkey = str(w)
    print(f'\n=== HR weight = {w} | AEL = {ael} ===')
    for seed in SENS_SEEDS:
        skey = str(seed)
        if skey in sens_progress[wkey]:
            r = sens_progress[wkey][skey]
            print(f'    seed={seed}: SKIPPED (already done) F1={r["f1"]:.4f} HR={r["hr"]:.4f}'
                  f'continue')
        model = _train_sens(ael, w, seed)
        f1, hr = _eval_sens(model)
        sens_progress[wkey][skey] = {'f1': f1, 'hr': hr}
        # Save immediately after each seed – crash-safe
        with open(PROGRESS_FILE, 'w') as _f:
            json.dump(sens_progress, _f, indent=2)
        print(f'    seed={seed}: F1={f1:.4f} HR={hr:.4f} [saved]')
        del model
        if DEVICE.type == 'mps': torch.mps.empty_cache()

sens_results = []
for w in SENS_HR_WEIGHTS:
    vals = sens_progress[str(w)]
    fls = [vals[str(s)]['f1'] for s in SENS_SEEDS]
    hrs = [vals[str(s)]['hr'] for s in SENS_SEEDS]
    sens_results.append({'w': w, 'f1_mean': np.mean(fls), 'f1_std': np.std(fls),
                        'hr_mean': np.mean(hrs), 'hr_std': np.std(hrs)})

print('\n===== SENSITIVITY RESULTS (mean ± std, N=3 seeds) =====')
print(f'{"w3":<6} {"Macro F1":>22} {"HR Recall":>22}')
print('-' * 52)
for r in sens_results:
    marker = ' ◀ (ours)' if r['w'] == 5.0 else ''
    print(f'{"{r["w"]}<6.1f} {"{r["f1_mean"]:.4f} ± {r["f1_std"]:.4f} {"{r["hr_mean"]:.4f} ± {r["hr_std"]:.4f}"'
          f' {marker}')

print('\n--- LaTeX rows (paste into Table 11) ---')

```

```

for r in sens_results:
    bold = r['w'] == 5.0
    tag = ' \\textbf{{ours}}' if bold else ''
    row = f'$\\mathbf{{{[1.0, 3.5, 3.0, {r["w"]:.1f}]}}}$ {tag}' if bold else f'
    fls = f'$\\mathbf{{{r["f1_mean"]:.4f} \\pm {r["f1_std"]:.4f}}}$' if bold
    hrs = f'$\\mathbf{{{r["hr_mean"]:.4f} \\pm {r["hr_std"]:.4f}}}$' if bold
    print(f'{row} & {fls} & {hrs} \\\\'

# Pick best weight by High-Risk Recall
best_entry = max(sens_results, key=lambda r: r['hr_mean'])
best_hr_weight = best_entry['w']
best_ael_weights = [1.0, 3.5, 3.0, best_hr_weight]
print(f'\n>>> Best HR weight by HR Recall: w={best_hr_weight}')
print(f'    AEL weights for ablation: {best_ael_weights}')
print(f'    HR Recall = {best_entry["hr_mean"]:.4f} ± {best_entry["hr_std"]:.4f}')
print('\nNow run Section 14 (Ablation) – it will use this weight automatically')
# — Save final results to CSV —
import pandas as _pd_sens
_sens_csv = os.path.join(PROJECT_DIR, 'ael_sensitivity_multiseed_results.csv')
_pd_sens.DataFrame(sens_results).rename(columns={
    'w': 'HR_weight', 'f1_mean': 'F1_mean', 'f1_std': 'F1_std',
    'hr_mean': 'HR_mean', 'hr_std': 'HR_std'}).to_csv(_sens_csv, index=False)
print(f'Saved \u2192 {_sens_csv}')

```

Resuming – 9 seed-runs already done.

```

=== HR weight = 3.0 | AEL = [1.0, 3.5, 3.0, 3.0] ===
seed=42: SKIPPED (already done) F1=0.7763 HR=0.9505
seed=0: SKIPPED (already done) F1=0.7872 HR=0.9598
seed=123: SKIPPED (already done) F1=0.7785 HR=0.9474

```

```

=== HR weight = 4.0 | AEL = [1.0, 3.5, 3.0, 4.0] ===
seed=42: SKIPPED (already done) F1=0.7790 HR=0.9628
seed=0: SKIPPED (already done) F1=0.7763 HR=0.9628
seed=123: SKIPPED (already done) F1=0.7855 HR=0.9628

```

```

=== HR weight = 5.0 | AEL = [1.0, 3.5, 3.0, 5.0] ===
seed=42: SKIPPED (already done) F1=0.7880 HR=0.9598
seed=0: SKIPPED (already done) F1=0.7802 HR=0.9536
seed=123: SKIPPED (already done) F1=0.7782 HR=0.9628

```

```

=== HR weight = 6.0 | AEL = [1.0, 3.5, 3.0, 6.0] ===

```

14. Multi-Seed Ablation — AEL vs CE vs Focal (Table 12)

Runs DenseNet-121 for 10 epochs under three loss functions (N=3 seeds each).
Run Section 13 (Sensitivity) first — it sets `best_hr_weight` used here for AEL weights.

```
In [ ]: import random, numpy as np, torch, torch.nn as nn, json, os
```

```

from torch.optim import AdamW
from torch.optim.lr_scheduler import CosineAnnealingLR
from sklearn.metrics import f1_score, recall_score

for _v in ['train_ldr', 'val_ldr', 'test_ldr']:
    if _v not in globals():
        raise RuntimeError(f'Run Sections 2 & 3 first ({_v} missing).')

ABLATION_SEEDS = [42, 0, 123]
ABLATION_EPOCHS = 25
PROGRESS_FILE = os.path.join(PROJECT_DIR, 'checkpoints', 'ablation_notebook_')
os.makedirs(os.path.dirname(PROGRESS_FILE), exist_ok=True)

# Load any previously saved progress so completed runs are skipped on resume
if os.path.exists(PROGRESS_FILE):
    with open(PROGRESS_FILE) as _f:
        _saved = json.load(_f)
    print(f'Resuming - {sum(len(v) for v in _saved.values())} seed-runs already done')
else:
    _saved = {}

def _set_seed(seed):
    random.seed(seed); np.random.seed(seed); torch.manual_seed(seed)
    if torch.cuda.is_available(): torch.cuda.manual_seed_all(seed)

def _run_one_ablation(loss_fn, seed):
    _set_seed(seed)
    model = build_densenet121().to(DEVICE)
    opt = AdamW(model.parameters(), lr=2e-4, weight_decay=1e-4)
    sch = CosineAnnealingLR(opt, T_max=ABLATION_EPOCHS, eta_min=1e-6)
    for ep in range(1, ABLATION_EPOCHS + 1):
        model.train()
        for imgs, labels in train_ldr:
            imgs, labels = imgs.to(DEVICE), labels.to(DEVICE)
            opt.zero_grad()
            loss = loss_fn(model(imgs), labels)
            loss.backward()
            nn.utils.clip_grad_norm_(model.parameters(), 1.0)
            opt.step()
        sch.step()
        print(f'  seed={seed}  epoch={ep}/{ABLATION_EPOCHS}', flush=True)
    model.eval()
    preds_all, trues_all = [], []
    with torch.no_grad():
        for imgs, labels in test_ldr:
            preds_all.extend(model(imgs.to(DEVICE)).argmax(1).cpu().numpy())
            trues_all.extend(labels.numpy())
    macro_f1 = float(f1_score(trues_all, preds_all, average='macro', zero_division=0))
    hr_recall = float(recall_score(trues_all, preds_all, average='macro', zero_division=0))
    [1 if t==3 else 0 for t in trues_all],
    [1 if p==3 else 0 for p in preds_all], zero_division=0))
    if DEVICE.type == 'mps': torch.mps.empty_cache()
    return macro_f1, hr_recall

```

```

# Use best weight from Section 14 sensitivity sweep; fallback to 5.0
_best_w = best_hr_weight if 'best_hr_weight' in globals() else 5.0
_ael_w = [1.0, 3.5, 3.0, _best_w]
print(f'Using AEL weights: {_ael_w} (run Section 13 Sensitivity first for aut

loss_configs = {
    'Cross-Entropy (uniform)': nn.CrossEntropyLoss(),
    'Focal Loss (gamma=2)': FocalLoss(gamma=2.0),
    'AEL (Ours)': AsymmetricEndoscopyLoss(weights=_ael_w),
}

ablation_results = {name: _saved.get(name, {}) for name in loss_configs}

for name, loss_fn in loss_configs.items():
    print(f'\n>>> {name}')
    for seed in ABLATION_SEEDS:
        key = str(seed)
        if key in ablation_results[name]:
            r = ablation_results[name][key]
            print(f' seed={seed}: SKIPPED (already done) F1={r["f1"]:.4f} HR=
                continue
        print(f' Running seed={seed}...')
        f1, hr = _run_one_ablation(loss_fn, seed)
        ablation_results[name][key] = {'f1': f1, 'hr': hr}
        # Save immediately after each seed – crash-safe
        with open(PROGRESS_FILE, 'w') as _f:
            json.dump(ablation_results, _f, indent=2)
        print(f' seed={seed}: F1={f1:.4f} HR={hr:.4f} [saved]')

print('\n===== ABLATION RESULTS (mean ± std, N=3 seeds) =====')
print(f'{"Loss":<44} {"Macro F1":>18} {"HR Recall":>20}')
print('-' * 84)
for name in loss_configs:
    vals = ablation_results[name]
    fls = [vals[str(s)]['f1'] for s in ABLATION_SEEDS]
    hrs = [vals[str(s)]['hr'] for s in ABLATION_SEEDS]
    print(f'{name:<44} {np.mean(fls):.4f} ± {np.std(fls):.4f} {np.mean(hrs):

print('\n--- LaTeX rows (paste into Table 12) ---')
for name in loss_configs:
    vals = ablation_results[name]
    fls = [vals[str(s)]['f1'] for s in ABLATION_SEEDS]
    hrs = [vals[str(s)]['hr'] for s in ABLATION_SEEDS]
    bold = 'AEL' in name
    fm = f'\mathbf{{{np.mean(fls):.4f}} \pm {np.std(fls):.4f}}' if bold el
    hm = f'{np.mean(hrs):.4f} \pm {np.std(hrs):.4f}'
    print(f'{name} & ${fm}$ & ${hm}$ \\\\'

# — Save final results to CSV —
import pandas as _pd_abl
_abl_rows = []
for _name in loss_configs:
    _vals = ablation_results[_name]

```

```

_fls = [_vals[str(s)]['f1'] for s in ABLATION_SEEDS]
_hrs = [_vals[str(s)]['hr'] for s in ABLATION_SEEDS]
_abl_rows.append({'loss': _name,
                  'F1_mean': float(np.mean(_fls)), 'F1_std': float(np.std(
                  'HR_mean': float(np.mean(_hrs)), 'HR_std': float(np.std(
_abl_csv = os.path.join(PROJECT_DIR, 'ablation_multiseed_results.csv')
_pd_abl.DataFrame(_abl_rows).to_csv(_abl_csv, index=False)
print(f'Saved \u2192 {_abl_csv}')

```

15. Model Calibration — Expected Calibration Error (ECE)

MC Dropout confidence scores drive the workload simulation threshold ($\tau=0.75$). This cell computes **Expected Calibration Error (ECE)** and plots **reliability diagrams** to verify that those scores are trustworthy.

- **ECE < 0.05** → well-calibrated (confidence $\approx$ accuracy)
- **ECE > 0.10** → consider temperature scaling before deployment

Run after Section 6 training and Section 8 evaluation are complete.

Requires `trained_models` and `test_ldr` in scope.

```

In [ ]: import numpy as np
import matplotlib.pyplot as plt
import torch
import torch.nn.functional as F

# — Prerequisites —
if 'trained_models' not in globals() or len(trained_models) == 0:
    raise RuntimeError('Run Section 6 (Training) first.')
if 'test_ldr' not in globals():
    raise RuntimeError('Run Section 3 (DataLoaders) first.')

def compute_ece(model, loader, n_bins=10):
    """Expected Calibration Error: measures confidence-accuracy alignment."""
    model.eval()
    all_confs, all_correct = [], []
    with torch.no_grad():
        for imgs, labels in loader:
            logits = model(imgs.to(DEVICE))
            probs = F.softmax(logits, dim=1).cpu().numpy()
            preds = probs.argmax(axis=1)
            confs = probs.max(axis=1)
            all_confs.extend(confs)
            all_correct.extend((preds == labels.numpy()).astype(float))
    all_confs = np.array(all_confs)
    all_correct = np.array(all_correct)

```

```

bins      = np.linspace(0, 1, n_bins + 1)
ece       = 0.0
bin_data  = []
for i in range(n_bins):
    mask = (all_confs > bins[i]) & (all_confs <= bins[i+1])
    if mask.sum() == 0:
        bin_data.append((0, 0, 0))
        continue
    acc = all_correct[mask].mean()
    conf = all_confs[mask].mean()
    ece += (mask.sum() / len(all_confs)) * abs(acc - conf)
    bin_data.append((conf, acc, mask.sum()))
return float(ece), bin_data

def plot_reliability(bin_data, model_name, ece, ax):
    """Reliability diagram: perfect calibration = diagonal line."""
    confs = [b[0] for b in bin_data if b[2] > 0]
    accs = [b[1] for b in bin_data if b[2] > 0]
    ax.bar(confs, accs, width=0.08, alpha=0.7, color='#3498DB', label='Model')
    ax.plot([0, 1], [0, 1], 'r--', linewidth=1.5, label='Perfect calibration')
    ax.set(xlabel='Confidence', ylabel='Accuracy',
           title=f'{model_name}\nECE = {ece:.4f}',
           xlim=(0, 1), ylim=(0, 1))
    ax.legend(fontsize=8); ax.grid(alpha=0.3)

fig, axes = plt.subplots(1, len(trained_models), figsize=(5 * len(trained_models), 5))
if len(trained_models) == 1:
    axes = [axes]

print('\n===== MODEL CALIBRATION (ECE) =====')
print(f'{"Model":<22} {"ECE":>8} {"Interpretation"}')
print('-' * 55)
for ax, (name, model) in zip(axes, trained_models.items()):
    ece, bin_data = compute_ece(model, test_ldr)
    interp = 'Well-calibrated' if ece < 0.05 else ('Moderate' if ece < 0.10 else 'Poor')
    print(f'{"name":<22} {"ece":>8.4f} {"interp}")')
    plot_reliability(bin_data, name, ece, ax)

plt.suptitle('Reliability Diagrams – Confidence vs Accuracy', fontsize=13, fontweight='bold')
plt.tight_layout()
plt.savefig(os.path.join(PROJECT_DIR, 'calibration_reliability.png'), dpi=150)
plt.show()
print('\nNote: ECE > 0.05 suggests temperature scaling before deployment.')
print('Reference: Guo et al. (2017) – On Calibration of Modern Neural Networks')

# — Save ECE values to JSON —————
import json as _json_ece
_ece_results = {}
for _name, _model in trained_models.items():
    _ece_val, _ = compute_ece(_model, test_ldr)
    _ece_results[_name] = round(_ece_val, 6)
_ece_path = os.path.join(PROJECT_DIR, 'ece_results.json')
with open(_ece_path, 'w') as _f:

```

```

    _json_ece.dump(_ece_results, _f, indent=2)
print(f'ECE values saved → {_ece_path}')
print(_ece_results)

```

16. Results Summary & Paper Tables

```

In [ ]: def print_paper_tables(results_dict: dict, ext_results: dict = None):
    print('\nTable 2. Per-Class F1 Score – All Models')
    print('=' * 95)
    print(f'{"Class":<22}', end='')
    for name in results_dict: print(f'{name:>18}', end='')
    print(f' Clinical Action')
    print('-' * 95)
    for i, (label, action) in enumerate(zip(LABEL_NAMES, RISK_ACTIONS)):
        print(f'{label:<22}', end='')
        for res in results_dict.values():
            print(f'{res["per_class"][i]:>18.4f}', end='')
        print(f' {action}')
    print('-' * 95)
    print(f'{"Macro F1":<22}', end='')
    for res in results_dict.values(): print(f'{res["macro_f1"]:>18.4f}', end='')
    print()
    print(f'{"High-Risk Recall":<22}', end='')
    for res in results_dict.values(): print(f'{res["high_risk_recall"]:>18.4f}')
    print()

    if ext_results:
        print('\nTable 3. Cross-Dataset Generalization (Kvasir-v2)')
        print('=' * 70)
        print(f'{"Dataset":<20}', end='')
        for name in results_dict: print(f'{name:>18}', end='')
        print()
        print('-' * 70)
        for site, res_by_model in ext_results.items():
            print(f'{site:<20}', end='')
            for name in results_dict:
                print(f'{res_by_model.get(name, {}).get("macro_f1", 0):>18.4f}')
            print()

    print('Results functions ready.')

```

```

In [ ]: def calibrate_thresholds(model, val_ldr, device):
    """Find per-class probability threshold that maximises macro F1 on validation
    model.eval()
    probs_all, trues_all = [], []
    with torch.no_grad():
        for imgs, labels in val_ldr:
            out = model(imgs.to(device))
            probs_all.extend(F.softmax(out, dim=1).cpu().numpy())
            trues_all.extend(labels.numpy())

```

```

probs_all = np.array(probs_all) # (N, 4)
trues_all = np.array(trues_all)

best_thresholds = np.zeros(NUM_CLASSES)
for c in range(NUM_CLASSES):
    best_f1, best_t = 0.0, 0.5
    for t in np.arange(0.1, 0.9, 0.02):
        preds = np.where(probs_all[:, c] >= t, c,
                        np.argmax(np.where(
                            np.arange(NUM_CLASSES) == c,
                            -np.inf, probs_all), axis=1))
        f1 = f1_score((trues_all == c).astype(int),
                      (preds == c).astype(int), zero_division=0)
        if f1 > best_f1:
            best_f1, best_t = f1, t
    best_thresholds[c] = best_t
    print(f' Class {c} ({LABEL_NAMES[c]:<18}): threshold={best_t:.2f} va
return best_thresholds

def predict_with_thresholds(model, loader, device, thresholds):
    """Predict using per-class calibrated thresholds instead of argmax."""
    model.eval()
    probs_all, trues_all = [], []
    with torch.no_grad():
        for imgs, labels in loader:
            out = model(imgs.to(device))
            probs_all.extend(F.softmax(out, dim=1).cpu().numpy())
            trues_all.extend(labels.numpy())
    probs_all = np.array(probs_all)

    # For each sample: pick the class whose probability most exceeds its thresh
    margins = probs_all - thresholds[np.newaxis, :]
    preds = np.argmax(margins, axis=1)

    macro_f1 = f1_score(trues_all, preds, average='macro', zero_division=0)
    print(f'\nWith calibrated thresholds - Macro F1: {macro_f1:.4f}')
    print(classification_report(trues_all, preds, target_names=LABEL_NAMES, ze
    return preds, macro_f1

# — Run calibration on best model —
if trained_models and 'results' in globals() and results and isinstance(list(r
    best_name = max(results, key=lambda n: results[n]['macro_f1'])
    best_model = trained_models[best_name]
    print(f'Calibrating thresholds for: {best_name}')
    thresholds = calibrate_thresholds(best_model, val_ldr, DEVICE)
    print(f'\nApplying to test set:')
    calibrated_preds, calibrated_f1 = predict_with_thresholds(
        best_model, test_ldr, DEVICE, thresholds)
else:
    print('[!] Train models first.')

```

Summary

Component	Detail
Task	4-class GI lesion risk stratification (ACG/ESGE guideline-aligned)
Models	DenseNet-121 · EfficientNet-B0 · DeiT-Tiny
Loss	Asymmetric Endoscopy Loss — weights [1.0, 3.5, 3.0, 5.0]
Training set	HyperKvasir — 21 of 23 finding classes → 4 risk tiers (7,633 images)
External validation	Kvasir-v2 zero-shot (8,000 images)
Explainability	GradCAM — mucosal/vascular feature localisation per risk class
Uncertainty	MC Dropout T=30, $\tau=0.75$ — escalates all Pre-malignant/High-Risk
Ablation	AEL vs Cross-Entropy vs Focal Loss (N=3 seeds, Table 12)
AEL sensitivity	HR weight sweep 3.0→7.0 (N=3 seeds, Table 11)
Best result	EfficientNet-B0: Macro F1=0.83 · Zero missed High-Risk · 43.9% auto-cleared

In []:

In []: